NIHR | National Institute for Health and Care Research

NIHR Programme Grants for Applied Research Programme

Scheme	Programme Grants for Applied Research
Funding Opportunity	2025/220 PGfAR Researcher-led

SUMMARY INFORMATION

Application Title	Treating auditory impairment to preserve cognition in people with mild cognitive impairment
Contracting Organisation	NORTH LONDON NHS FOUNDATION TRUST
Contracting Organisation Department	Services for Ageing and Mental Health
Start Date	01/04/2026
End Date	30/09/2031
Duration (months)	66
Total Cost to NIHR	£3,420,354.46
NHS Support Costs	£10,822.98
NHS Excess Treatment Costs	£88,594.08
Non-NHS Excess Treatment Costs	£0.00
Has this application been previously submitted to this or any other funding body?	No

LEAD APPLICANT & RESEARCH TEAM

Lead Applicant

NIHR Programme Grants for Applied Research Programme

Full Name	Professor Sergi Costafreda Gonzalez
Organisation	University College London
Department	Mental Health of Older People
Proposed Role	I led the TACT pilot that proved supported hearing aids work for people with cognitive impairment: 75% uptake versus 22% in standard audiological care. I am positioned to adapt this innovation for primary care delivery and test its cognitive impact at scale through TACT+. As Professor of Psychiatry of Ageing at UCL and consultant at Islington Memory Service, I bridge research and clinical reality. I've spent a decade treating MCI while building research partnerships with GP practices. I currently direct ENHANCE (£2.9m NIHR programme), successfully recruiting healthy older adults through primary care networks—demonstrating my ability to engage GPs and navigate primary care pathways directly relevant to TACT+. Through ENHANCE, I've learned to design interventions that work when delivered by non-specialists, create training that builds confidence in primary care teams, and develop digital tools that work for older people and support busy practices. My £8m research portfolio spans sensory health, dementia prevention, technology and implementation science, generating 27,000 citations and shaping international policy through the Lancet Dementia Commission. I will reduce clinical sessions from 3 to 2 weekly to provide hands-on leadership across all five workstreams. My co-investigators bring multidisciplinary expertise in hearing research, primary care, implementation and trial operational excellence. TACT+ represents the convergence of my clinical expertise in cognitive ageing, proven ability to deliver complex interventions through primary care, and vision for accessible dementia prevention. I have the experience, relationships, and team to make this work in the real NHS.
ORCID	0000-0001-6914-086X
% FTE Commitment	17%

NIHR Programme Grants for Applied Research Programme

Co-Applicants

Full Name	Dr Ruan-Ching Yu
Organisation	University College London
Department	Mental Health of Older People
Proposed Role	Dr Ruan-Ching Yu brings unique qualifications to lead TACT+ programme management. As first author on the TACT pilot trial publication (Yu et al., 2025) and of the systematic review establishing hearing-dementia links (Yu et al., 2024), and with a PhD on cultural adaptation and validation of cognitive testing for MCI, she possesses deep knowledge of both the intervention and its evidence base. Her coordination of TACT pilot and this complex NIHR application shows her organisational capability. She has assembled a multidisciplinary research team, worked with PPI, navigated NHS governance structures, engaged Primary Care and RDNs, and built enthusiasm across diverse stakeholders. This achievement reveals her ability to translate between clinical, academic, and operational worlds. While new to formal programme management, Dr Yu will be supported by UCL CCTU's dedicated trial coordinator and team, ensuring operational excellence. This structure allows her to focus on scientific leadership while developing PM expertise through mentored practice. As Programme Manager reporting to the CI, Dr Yu will transform strategic vision into executable plans. She will coordinate across five workstreams and multiple sites, managing day-to-day delivery and budget oversight. Her responsibilities span co-production workshops (WS1), field-testing logistics (WS2), trial operations partnership with CCTU (WS3), and implementation outputs (WS4-5). Dr Yu's scientific expertise in hearing-cognition pathways ensures clinical relevance drives operational decisions. Her proven ability to build consensus among diverse stakeholders provides the relationship skills essential for multi-site coordination.
ORCID	0000-0002-8081-5300
% FTE Commitment	100.00%

NIHR Programme Grants for Applied Research Programme

Full Name	Professor Anne GM Schilder
Organisation	University College London
Department	Ear Institute
Proposed Role	Prof Anne Schilder is a Consultant ENT surgeon at UCLH and trialist, NIHR Senior Investigator and Director of the NIHR UCLH BRC Hearing Health Theme, where she uses qualitative and quantitative research methods to establish how best to educate ear and hearing care professionals about the role of cognition in hearing care for older adults, how to detect early cognitive impairment in ear and hearing clinics for older adults and provide tailored care to those identified with early cognitive impairment (Omar et al., 2023; Yu et al., 2025). As the past NIHR CRN ENT National Specialty Lead she has a large network across specialties involved in the proposed TACT+ programme and in-depth knowledge of research delivery within the NHS setting. She will provide senior clinical academic guidance throughout the programme and work with ENT and Audiology Specialties to ensure the trial intervention is robust and practically implementable within the NHS. Her experience in establishing best practice in hearing care will be vital in navigating the novel governance and clinical safety challenges of implementing a supported self-fitting pathway in a non-specialist primary care setting. She will oversee and actively participate in the co-production of the TACT+ pathway and its field testing (WS1, WS2). She will provide expert input in the definitive clinical trial protocol and advise on trial set-up and delivery (WS3), as well as the development of the implementation toolkit and dissemination strategy (WS4, WS5)
ORCID	0000-0002-5496-4580
% FTE Commitment	7.50%

NIHR Programme Grants for Applied Research Programme

Full Name	Dr Ross Dunne
Organisation	GREATER MANCHESTER MENTAL HEALTH NHS FOUNDATION TRUST
Department	Later Life Psychiatry
Proposed Role	Ross Dunne is a Consultant Psychiatrist and Director of the Greater Manchester Dementia Research Centre. He led development of the "Mild Cognitive Impairment: the Manchester consensus" (Dunne et al., 2021), which has transformed clinical practice across the UK. This highly-cited publication provided the first UK expert consensus on diagnosing and managing MCI, standardising practice where formal guidance was lacking. Within TACT+, Dr Dunne will provide senior clinical oversight for cognitive eligibility criteria and lead the Manchester field-testing site. He will establish primary care partnerships and operational infrastructure for WS2 delivery, leveraging his local relationships to ensure efficient site set-up. His focus on equitable recruitment will help engage diverse Greater Manchester communities while ensuring trial governance meets NHS standards across mental health and primary care settings. He will support co-production of the TACT+ pathway, developing clinically grounded inclusion criteria (WS1), lead Manchester field-testing operations (WS2), co-develop the trial protocol for identifying potential MCI cases in primary care (WS3), and contribute to interpretation and dissemination of findings (WS5). His MCI expertise and established Manchester partnerships ensure TACT+ can rapidly mobilise in the region while reaching communities often underrepresented in research.
ORCID	0000-0003-2046-6193
% FTE Commitment	3.10%

NIHR Programme Grants for Applied Research Programme

Full Name	Dr Helen Henshaw
Organisation	University of Nottingham
Department	NIHR Nottingham Biomedical Research Centre/Hearing Sciences, Mental Health & Clinical Neurosciences
Proposed Role	Dr Helen Henshaw is a Cognitive Psychologist and Nottingham BRC Principal Research Fellow. She leads a programme of research in auditory rehabilitation and cognitive training, with expertise in promoting hearing aid use. As Chair of the British Society of Audiology's Cognition in Hearing Special Interest Group, she bridges hearing and cognitive sciences. She recently led the NIHR James Lind Alliance Priority Setting Partnership for Dementia and Hearing Conditions, bringing patients, carers, and clinicians together to identify the field's top research priorities. Dr Henshaw will apply her behavioural science expertise to design the TACT+ support package, ensuring it maximises engagement and adherence among people with cognitive impairment. She will lead development of the hearing coach manual and hearing rehabilitation components of the TACT+ app during co-production (WS1). She will co-lead the optimisation and field-testing phase (WS2) and lead evaluation of intervention components within the process evaluation (WS4). Her unique combination of cognitive psychology, hearing rehabilitation expertise, and patient priority insights ensures TACT+ interventions are both evidence-based and aligned with user needs.
ORCID	0000-0002-0547-4403
% FTE Commitment	10.00%

NIHR Programme Grants for Applied Research Programme

Full Name	Ms Fahima Khatun
Organisation	University College London
Department	Ear Institute
Proposed Role	Fahima Khatun is PPIE and EDI Lead of the NIHR UCLH BRC Hearing Health Theme, where she leads a diverse panel of public contributors with lived experience of hearing conditions. As Director of a home care company in East London, she has extensive first-hand expertise supporting older adults with cognitive impairment and working with underserved communities across healthcare settings. This dual perspective gives her unique insight into the health inequalities and access barriers TACT+ aims to address. Ms Khatun will lead the programme's integrated PPIE and EDI strategy. She will ensure meaningful involvement is embedded throughout, grounded in established frameworks (UK Standards for Public Involvement, 4Pi). She will co-design and facilitate inclusive co-production workshops (WS1), develop the EDI strategy to support recruitment from underserved communities (WS3), co-chair the PPIE panel to shape research throughout (WS1-5), and ensure dissemination outputs are co-produced and accessible for people with hearing loss and cognitive difficulties (WS5). Her unique combination of academic PPIE leadership and daily experience caring for older adults with cognitive impairment ensures TACT+ remains grounded in real-world needs while advancing equity in dementia prevention research.
ORCID	0009-0001-6135-8765
% FTE Commitment	5.31%

NIHR Programme Grants for Applied Research Programme

Full Name	Dr Liz Steed
Organisation	Queen Mary University of London
Department	Centre for Primary Care, Wolfson Institute of Population Health
Proposed Role	Dr Liz Steed is Reader in Health Psychology and an expert in behaviour change and implementation science. She has extensive experience developing theory-based interventions for successful trials (STAMINA, RP-PG-1016-2007) and leading process evaluations in implementation trials (IMP2ART RRP-PG-1016-20008, PETAL). As implementation lead for ENHANCE, she has established strong working relationships with the current team. Internationally, she teaches behaviour change in pulmonary rehabilitation and is an active member of the European Respiratory Society's Psychology and Behavioural Scientists division. Dr Steed will lead TACT+'s process evaluation and implementation strategy, ensuring the pathway is grounded in behavioural and implementation theory while generating deep understanding of how it can be adopted and sustained across diverse NHS primary care contexts. She will co-lead co-production using the COM-B model and Theoretical Domains Framework (WS1). Her leadership will be central to WS4, where she will conduct mixed-methods process evaluation using Normalisation Process Theory and develop the theory-informed implementation toolkit for NHS adoption. Her expertise ensures TACT+ moves beyond efficacy to real-world implementation from the outset.
ORCID	0000-0003-1926-3196
% FTE Commitment	10.00%

Full Name	Ms Felicia Ikeji
Organisation	University College London
Department	Comprehensive Clinical Trials Unit
Proposed Role	Felicia Ikeji is Head of Clinical Trial Operations at UCL CCTU. She will provide senior operational oversight for the full-size multi-site RCT (WS3), ensuring delivery to the highest methodological and regulatory standards. Her expertise in troubleshooting common trial delays will be invaluable in mitigating risks and ensuring efficient delivery of high-quality results on time and budget. Ms Ikeji will oversee the CCTU team responsible for day-to-day trial management, including protocol development, ethics and regulatory approvals, site set-up, and implementation of robust recruitment and data management strategies. She will ensure that all trial operations maintain GCP compliance throughout. Her leadership ensures TACT+ benefits from UCL CCTU's established infrastructure and proven track record in operational excellence in delivering complex multi-site trials.
ORCID	0009-0001-8734-445X
% FTE Commitment	10.00%

NIHR Programme Grants for Applied Research Programme

Full Name	Dr Devina Maru
Organisation	University College London
Department	Research Department of Primary Care and Population Health
Proposed Role	Dr Devina Maru is a practising GP with ENT special interest, Co-Deputy Director for Medical Education in Primary Care at UCL, and locum GP at Islington iHUB Enhanced Access Service. She shaped the RCGP Hearing Loss Education Toolkit, led a £169K Regulators Pioneers Fund project on health inequalities, and advised the NIHR JLA Priority Setting Partnership for Dementia and Hearing Conditions. As former RCGP National Clinical Champion for Hearing Loss and current Primary Care representative on the NIHR UCLH BRC Hearing Health Theme Board, she brings deep experience converting national policy into practical, equitable primary care services. In WS1, Dr Maru will steer co-production of the hearing pathway to dovetail with GP workflows, digital priorities and ICB transformation agendas, advising on EMIS/SystmOne search protocols and practice-nurse scripts. During WS2 field-testing, she will address usability or equity barriers emerging in practices. In WS3, she will advise on clinician and patient engagement strategies for inclusive recruitment across diverse communities. In WS4, she will co-lead development of the implementation toolkit and CPD resources, ensuring the hearing-coach model is ready for scalable primary care adoption. Her dual perspective as frontline GP and policy leader ensures TACT+ remains grounded in primary care reality while advancing system-wide change.
ORCID	0000-0003-4344-4069
% FTE Commitment	10.00%

NIHR Programme Grants for Applied Research Programme

Full Name	Ms Monica Panca
Organisation	University College London
Department	Comprehensive Clinical Trial Unit
Proposed Role	Monica Panca is UCL CCTU Senior Health Economist with expertise in economic evaluation within clinical trials. She has extensive experience conducting economic evaluations (within-trial and decision modelling) across different disease areas, including cognitive impairment. Ms Panca will lead the economic evaluation of TACT+, providing robust evidence on value for money and affordability to inform NHS commissioners and policymakers, supporting real-world uptake if the intervention proves effective. Within the main trial, she will oversee development of relevant CRFs, protocol and health economic analysis plan. She will conduct cost-effectiveness analyses from NHS/Personal Social Services and societal perspectives, using trial data on resource use, intervention costs, and quality of life. She will also lead long-term modelling to project NHS implications beyond the trial timeframe and budget impact analysis for national rollout (WS3). Her expertise ensures TACT+ generates the economic evidence essential for commissioning decisions and implementation planning.
ORCID	0000-0002-4031-6478
% FTE Commitment	3.00%

NIHR Programme Grants for Applied Research Programme

Full Name	Dr Hannah Semeraro
Organisation	RNID
Department	Insight and Policy
Proposed Role	Dr Hannah Semeraro is an Audiologist and Head of Insight and Evidence at RNID, the national charity supporting people who are deaf, have hearing loss, or tinnitus. Her PhD focused on developing novel tools to assess functional hearing ability, and her clinical and teaching experience includes training clinical audiologists. As Head of Insight and Evidence at RNID, Dr Semeraro brings subject-matter expertise and connects TACT+ to the communities RNID supports. She will ensure the intervention is patient-centred, equitable, and informed by lived experience. RNID's extensive networks and insights into barriers faced by people with hearing loss will strengthen the design, delivery, and dissemination of TACT+. Dr Semeraro will support co-production of the TACT+ pathway and co-lead development of the training package for primary care staff, drawing on her audiology teaching experience (WS1). She will co-lead public and third-sector engagement activities, ensuring outputs are accessible and inclusive (WS5). RNID will publicise the trial through its communication channels, support recruitment via its networks, and help disseminate findings. RNID's policymaker relationships will support translation of TACT+ findings into national policy recommendations (WS5). Her dual role as clinical expert and charity leader ensures TACT+ bridges research excellence with real-world impact for people with hearing loss.
ORCID	0009-0007-9871-4843
% FTE Commitment	3.44%

NIHR Programme Grants for Applied Research Programme

Full Name	Dr Gareth Smith
Organisation	MID AND SOUTH ESSEX NHS FOUNDATION TRUST
Department	Audiology
Proposed Role	Gareth Smith is Clinician Scientist in Audiology, Head of Audiological Science and Clinical Lead at Mid and South Essex NHS Foundation Trust. As former President of the British Academy of Audiology and current Audiology Editor for ENT & Audiology News, his expertise lies in strategic conception and operational delivery of advanced clinical audiology services within the NHS. In TACT+, Mr Smith will lead technical and clinical development of the hearing intervention. Crucially, he will guide selection of appropriate self-fitting hearing aids for our MCI population, drawing on his deep knowledge of the rapidly evolving OTC market and clinical evidence base. His frontline NHS experience ensures we choose devices that balance innovation with practical usability for cognitively vulnerable users. He will co-create protocols converting Tympa diagnostic data into safe fitting parameters and remote validation dashboards operable by primary care staff, ensuring the pathway is clinically sound and implementable within NHS realities. He will co-lead development and operationalisation of the pathway (WS1-2) and provide expert guidance on NHS device procurement and data tracking (WS4). His clinical acumen and finger on the pulse of hearing technology ensures TACT+ selects and implements solutions that work in real-world practice.
ORCID	0000-0002-4724-9589
% FTE Commitment	5.00%

Full Name	Professor Nick Freemantle
Organisation	Comprehensive Clinical Trials Unit at UCL
Department	Institute for Clinical Trials and Methodology
Proposed Role	Prof Nick Freemantle is UCL CCTU Director, Professor of Clinical Epidemiology & Biostatistics, and a leading trial methodologist. He will provide senior oversight for the programme's entire methodological design, ensuring the highest standards of rigour. As ultimate guarantor of the trial's robustness and credibility, he will ensure the design directly addresses funding committee feedback. His leadership will focus on the definitive RCT (WS3). He will direct the statistical analysis plan, ensuring alignment with CONSORT-Equity extension to investigate effects in diverse subgroups, and provide expert guidance to the independent oversight committees (TSC/DMEC). Prof Freemantle's involvement ensures TACT+ meets the gold standard for clinical trial conduct and produces robust evidence, fit for NHS decision-making and change of practice.
ORCID	0000-0001-5807-5740
% FTE Commitment	7.50%

NIHR Programme Grants for Applied Research Programme

Full Name	Professor Gill Livingston
Organisation	University College London
Department	Mental Health of Older People
Proposed Role	Prof Gill Livingston is a Consultant Psychiatrist and world-leading expert in dementia prevention, serving as lead author of the Lancet Dementia Commission. An experienced trialist in cognitive impairment and dementia, she brings deep expertise in participant recruitment, retention, and carer engagement. She will provide senior clinical and academic guidance throughout TACT+, advising on strategies to ensure patient acceptability and trial success. As named lead for Research Capacity Building, she will oversee development of our early career researchers. Her expertise will be key for interpreting trial findings in the context of national and international dementia strategy. Her proven dissemination skills, evidenced by the global impact of the Lancet Commission, will help translate TACT+ findings into policy and practice change (WS5). She will oversee two PhD studentships and two research associate posts, ensuring the programme builds lasting capacity in hearing and dementia research. Prof Livingston's leadership ensures TACT+ delivers both rigorous evidence and real-world impact in dementia prevention.
ORCID	0000-0001-6741-5516
% FTE Commitment	5.00%

NIHR Programme Grants for Applied Research Programme

Full Name	Dr Nishchay Mehta
Organisation	UNIVERSITY COLLEGE LONDON HOSPITALS NHS FOUNDATION TRUST
Department	Royal National ENT and Eastman Dental Hospitals
Proposed Role	Prof Nishchay Mehta is Consultant ENT surgeon at UCLH, Chair of Otology and Auditory Implants, AI Scientist and Co-director of the NIHR UCLH BRC Hearing Health Theme. He leads data-driven research applying AI to routine NHS data to understand hearing loss phenotypes and develop targeted treatments. He co-leads the NIHR I4I grant developing soft robotics for inner ear therapeutics. Critically for TACT+, he led development and rollout of remote ear and hearing care pathways in North London PCNs using Tympa technology, and worked with NHSE to expand this nationally to 350+ community pharmacies. This same Tympa audiometry technology will guide self-fitting of hearing aids in TACT+. Together with co-applicants representing Audiology, Dementia, and Primary Care, he will co-lead design and operationalisation of the TACT+ pathway (WS1-2), advise on pragmatic workflows, training and governance of hearing coaches (WS2-3), and guide integration of the TACT+ app into NHS pathways (WS4-5). His unique combination of clinical expertise, NHS systems knowledge, and proven ability to scale digital hearing care nationally positions him to ensure TACT+ delivers practical, implementable solutions.
ORCID	0000-0001-9317-3055
% FTE Commitment	5.00%

NIHR Programme Grants for Applied Research Programme

Full Name	Mr Terry Emery
Organisation	NIHR University College London Hospitals Biomedical Research Centre
Department	Hearing Health theme
Proposed Role	Terry Emery is a former freelance copywriter and public representative within the NIHR UCLH BRC Hearing Health Theme PPIE panel, bringing personal lived experience of age-related hearing loss to research. He has contributed to major initiatives including the James Lind Alliance Priority Setting Partnership, which identified the top 10 research priorities for mild-to-moderate hearing loss, and serves as lay member of the DHSC/NHS England working group that co-produced the national Hearing Loss & Tinnitus Research Action Plan, translating policy into practice. As public co-applicant on TACT+, Mr Emery will sit on the Programme Management Group and Trial Steering Committee, ensuring lived-experience insights shape governance decisions, recruitment strategy and participant retention. Crucially, he will guide co-development of the intervention and its digital components, advising how to make them accessible for people with memory or hearing difficulties. He will co-produce the TACT+ pathway and participant-facing materials, ensuring plain-language clarity (WS1). Together with co-applicant Linda Kirby, he will co-lead the lay dissemination plan—developing plain-English summaries, identifying community channels and creating accessible formats—ensuring findings reach the broadest public audience (WS5). His professional communication skills combined with personal hearing loss experience uniquely position him to bridge research and public understanding.
ORCID	0009-0004-6626-4567
% FTE Commitment	10.00%

NIHR Programme Grants for Applied Research Programme

Full Name	Ms Linda Kirby
Organisation	NIHR University College London Hospitals Biomedical Research Centre
Department	Hearing Health Theme
Proposed Role	Linda Kirby is a retired university library assistant and long-standing public representative within the NIHR UCLH BRC Hearing Health Theme PPIE panel. Diagnosed with congenital hearing loss in mid-life, she combines lived experience with professional expertise in plain-language communication and inclusive resource curation, skills that align directly with TACT+'s aims to create accessible, patient-centred information. As public co-applicant on TACT+, Ms Kirby will hold a seat on the Programme Management Group and Trial Steering Committee, providing ongoing scrutiny of recruitment and retention plans. Alongside Terry Emery, she will co-design the pathway and digital interface, advising on layout, navigation and multimedia cues that work for people with hearing and memory challenges. She will conduct detailed readability and usability reviews of study documents and app screens (WS1), and co-lead the lay dissemination strategy to ensure results are shared in meaningful, accessible ways (WS5). Her unique combination of information science expertise and personal hearing loss experience ensures TACT+ materials truly serve those who need them most.
ORCID	
% FTE Commitment	10.00%

NIHR Programme Grants for Applied Research Programme

APPLICATION DETAILS

Scientific Abstract

Research Question

Is "TACT+"—a proactive, primary care embedded hearing pathway combining handheld audiometry, self-fitting hearing aids and remote coaching—effective and cost effective in slowing cognitive decline for adults ≥55 with mild cognitive impairment (MCI) and hearing loss versus usual care?

Background

MCI affects up to one in five older adults, with 14 to 23% progressing to dementia every 18 months. At the same time, 25% of UK adults aged 50–89 have hearing loss—a key modifiable dementia risk. Untreated hearing loss increases cognitive load, accelerates neurodegeneration and fuels social isolation, yet fewer than 15% of those who could benefit use hearing aids. In our TACT pilot, run in memory services, personalised support more than tripled regular aid use (22→75%). The ACHIEVE trial reported a 62% reduction in cognitive decline among high-risk groups, comparable to our target population.

We will adapt the intervention for Primary Care “hearing–cognition hubs” using handheld Tympa testing and self-fitting hearing aids, supported by a coach and an app. The app enables goal-setting, rehabilitation games, reminders and secure coach messaging, and coaches guide fitting, troubleshoot issues and build confidence through remote sessions.

Aims

Co-develop, evaluate and enable national roll out of TACT+ to preserve cognition, reduce inequalities and improve quality of life for people with MCI and hearing loss.

Objectives

1. Co-design the TACT+ intervention, coaching, app and training (WS1).
2. Test feasibility in two PCN hubs (WS2).
3. Establish clinical and economic effectiveness in a large-scale RCT (WS3).
4. Conduct a mixed methods implementation study and NHS-ready toolkit (WS4).
5. Dissemination of results to influence policy, practice and public uptake, focusing on underserved groups (WS5).

Methods

WS1 Co design (M1–12): co- workshops apply the Behaviour Change Wheel to co-create Tympa assessment, immediate self-fit of hearing aids, app coaching and NHS audiology links.
WS2 Field Testing (M7–18): Two contrasting hubs (London, Manchester) enrol 62 participants to test recruitment, workflows, feasibility and equity strategies.
WS3 RCT (M16–63): Two arm RCT with masked assessors (n = 424, 12 sites), including a 6-month internal pilot. Eligibility: 55 y, MoCA 18–25 (screening rather than formal MCI diagnosis), symmetrical 30–60 dB loss. Intervention: TACT+; Care as usual: audiometry feedback with advice to consult GP. Primary outcome: MoCA change at 18 months; secondary outcomes: cost-effectiveness, dementia incidence, hearing outcomes, mood, social function.
WS4 Implementation (M1–63): Normalisation Process Theory guided evaluation with fidelity metrics and >50 stakeholder interviews to identify barriers, enablers and core-plus-adaptable elements; outputs include commissioning briefs, training resources and budget tools.
WS5 Dissemination (M36–63): Co created lay summaries, policy briefs, CPD resources and open-source code; partnerships with RNID, NIHR Evidence and third sector groups maximise reach into deprived and ethnic minority communities.

Impact
If effective, TACT+ will offer a low cost, digitally enabled route to preserve cognition, halve progression to dementia, and reduce inequalities by reaching high deprivation PCNs. Outputs - open access toolkit, commissioning brief, and co-developed PPIE resources—will position the NHS to roll out hearing care prevention at scale within six years.

Plain English Summary
As we get older, some of us start to have trouble with memory or thinking clearly. When these changes are noticeable but people can still manage daily life, doctors may call it mild cognitive impairment (MCI). About one in six people over 60 have MCI, and about 15% of them develop dementia every 18 months.

Hearing loss is also common in later life. It can make everyday life harder — conversations become difficult, people withdraw socially, and loneliness follows. Hearing loss can make memory worse. The brain has to work harder to hear, and the parts that process sound shrink.

Hearing aids help people stay connected, feel better, and live more independently. They may also help protect thinking and memory. But many people with memory problems don't get hearing aids, or struggle to use them.

In a small study with 58 people who had both MCI and hearing loss, we tested a new approach: doing hearing tests and fitting hearing aids right away, with three months of support from a hearing coach. The results were encouraging:
* 75% used their hearing aids regularly (compared to only 22% with usual care)
* People did better on memory tests and felt better, although a larger study is needed to confirm this

We now plan a bigger study. To find people earlier, we will work in GP practices, where mild memory difficulties often first appear.

Half the participants will get hearing aids they fit themselves, with help from a hearing coach. The coach will help them get used to the aids, fix problems, and keep using them. People will also get an app with reminders and a way to contact the coach. The goal is better hearing, mood, social life, and quality of life. This may delay more serious memory problems like dementia.

The other half will receive usual care: a letter with their hearing test results and advice to speak to their GP about hearing referral.

After 18 months, we will compare memory, hearing aid use, mood, social links, quality of life, and costs between the two groups.

The whole project will take five years. The first 18 months will focus on setting up two hubs in Manchester and London for early testing. Local practices and memory services will refer people there. The next three and a half years will involve running the full trial and analysing the results.

Alongside the trial, we'll also look at how this approach works in real life. We'll try it out in NHS Neighbourhood Health Centres and GP practices, and talk to patients and staff about their experiences. What we learn will help us make it easier for the NHS to use this approach in the future.

People and Communities:

Our lived experience panel has shaped this project from the start and will continue to be involved. They will co-design the intervention, help with study materials and recruitment and guide how we share results and plan for future use in the NHS.

We are committed to including people from all backgrounds, especially those who are often left out of

research. Our team has a strong track record of working with diverse communities, so the findings can be useful for everyone.

Sharing the Results:

We will share what we learn with the public, national charities, and patient groups — who will also help us spread the word. Together, we'll share updates through a website, social media, and newsletters, and we'll present at public events and publish in journals. If it works well, we'll use what we've learned to help the NHS offer this approach more widely.

Changes from Previous Stage

1. Some intervention development and piloting already done - value for money?
Our Stage 2 request is £3.42M (£50K less than Stage 1) despite expanded scope and ambition. We have shortened the overall duration from 72 to 66 months by shifting recruitment from memory services to higher-throughput primary care hubs. Our previous pilot work showed high hearing aid use with support, but further development is needed to scale the intervention to adapt it for mostly remote rather than fully face-to-face support and for deployment in primary care.

2. Recruiting from memory clinics intervenes too late
We agree. The revised model now identifies and supports people with MCI earlier in primary care, requiring three key changes:
- Intervention: We moved from audiologist-led fitting which would not be scalable in primary care, to a supported self-fitting model leveraging technology and using CE/UKCA-marked OTC hearing aids and Tympa for assessment, all delivered by trained non-specialists with remote audiology input which is deliverable in primary care and aligned with direction of NHS policy.
- Primary Outcome: We replaced the ACE-III with the Montreal Cognitive Assessment (MoCA) a briefer, well-validated tool requiring less training and specialised staff and feasible for primary care delivery.
- Team: We increased FTEs for Drs Maru (primary care), Henshaw (behavioural science), and Steed (implementation) to reflect expanded roles in design and delivery.

3. Why is a trial needed
We need an RCT to determine if supported hearing aid use results in meaningful cognitive benefit and likely to lead to dementia prevention, as this is the likely 'game-changer' in terms of patient benefit and costs that can create the drive to improve the current care as usual (CAU) model for hearing care that fails most people with MCI. We also need a trial of TACT+ vs CAU to determine broader and important patient benefits, on mood, social interaction and quality of life. Finally we need a trial to show if TACT+ is cost-effective relative to CAU. Full rationale with references is in p4 of the Scientific case.

4. Primary Outcome Timing
Eighteen months balances detection of cognitive change and feasibility. MoCA is sensitive within this timeframe in converters vs stable MCI. Hearing aid habit formation consolidates in 6–12 months so benefits of sustained use on cognition and other outcomes expected within this window. Readouts at 6 months (resource use via CSRI) and 12 months (all primary and secondary outcomes) will reduce recall bias and strengthen imputation if is required. See p.16.

5. Effect size Justification and Sample Size
MoCA has strong evidence of sensitivity and MCID in MCI populations. A 1.5-point effect is conservative and meaningful, endorsed by our PPIE contributors: MoCA has lower burden, and interpretability on likely impact on daily life. Despite switching outcomes, our power calculations remain valid and the sample size (n=424) unchanged under conservative assumptions. See p.17.

6. Inclusion of under-served populations
We will recruit from high-deprivation areas (IMD 1–4) using GP searches, SMS, and opportunistic routes. We provide 4G tablets and coaching to reduce digital exclusion. Materials will be co-produced

with lived experience contributors for cultural and digital accessibility. WS4 oversamples INCLUDE groups and explores contextual factors. The main trial will analyse outcomes by IMD, ethnicity, and age. See pp. 9, 16, 19, 22–25.

7. Role of PPIE
PPIE shaped this application and will shape every workstream. Focus groups informed design, outcome selection, how to sensitively communicate assessment results to participants, benefits of hearing aids beyond cognition, how to approach coaching, and app simplification. Two public contributors became co-applicants and will continue to guide design, delivery, and governance. A PPIE co-investigator leads involvement; a co-production group will support implementation and materials. Support and evaluation structures are in place. See pp. 6–7, 24–25.

8. Plain English summary was excellent
Thank you, we have updated it.

9. Capacity building
We removed the MSc post and now support two PhDs (behaviour change, audiology) and two pre-doctoral RAs (audiology, implementation). Trial staff, including the research audiologist, will receive protected time, mentorship, and training. This aligns with PGfAR and NIHR Academy guidance and represents 19% of the budget. Prof Livingston is the Named Training Contact. See pp. 26–27.

10. Timeline and regulatory risks
We revised the timeline for realism, in consultation with our CTU: reduced total duration, but lengthened the preparatory phase for hiring and approvals. We do not anticipate needing MHRA approval as all devices used will hold CE/UKCA marks and be deployed within intended use. Tympa is CE-marked. TACT+ app is not a regulated medical device. See pp. 6–7, 15–18.

11. Liaising with RDNs.
We developed this bid with our local RDN, NOCLOR and their Primary Care Research Delivery team, with whom we co-run ENHANCE where we have started recruitment in GP practices in London, Yorkshire and Humber, and have more areas ready (NIHR203670). For TACT+ we have engaged RDNs in the North West, North East, Midlands, East, South Coast, and Wales to support site identification and recruitment. See p. 10.

12. Open Access costs
We have not included open access costs.

Additional clarifications from referees
- Carers: Most participants will not have carers, but where relevant, costs will be included in the economic evaluation.
- Health utility: We selected HUI3 due to its sensitivity to changes in cognition and hearing loss, supported by studies in MCI and audiology populations.
- Implementation: WS4 now explicitly evaluates feasibility, fidelity, and adoption using NPT and Proctor's Outcomes and includes a PhD and RA post to embed implementation capacity and generate an NHS-ready toolkit. See pp. 22–23.

Working with People and Communities Summary
Leadership Through Lived Experience
Our public involvement is co-led by Linda Kirby and Terry Emery, who bring complementary perspectives on hearing and cognition as co-applicants. Linda, a retired university library assistant diagnosed with congenital hearing loss in mid-life, combines professional expertise in clear communication with personal insight into late-diagnosis challenges. Terry, a former copywriter who contributed to the James Lind Alliance Priority Setting Partnership on hearing and dementia, brings his

perspective on how hearing loss intersects with cognitive concerns.
Supporting them is Fahima Begum, our PPIE lead with over a decade of experience facilitating inclusive research. She brings professional expertise and personal experience of caring for people with cognitive impairment and hearing loss. This PPIE team works closely with Hannah Semeraro, RNID Head of Insight and Evidence and audiologist.

Building an Inclusive Partnership
We will proactively recruit a diverse PPIE panel. Our commitment to inclusion means:
• Flexible participation: Hybrid meetings, with options for in-person, video, or phone involvement.
• Accessible formats: Large print, Easy Read materials, hearing loops, and live captioning as standard
• Financial support: we will fund costs of travel, caring needs, technology, training, and other participation costs as required
• Tailored engagement: Members can contribute in ways that suit their preferences through in face to face or remote meetings, written or oral contributions.
We will track our reach using NIHR Research Inclusion tools to ensure we are hearing from under-represented voices.

Co-Production Across All WS
WS1 - Design: PPIE partners will co-create the intervention with the rest of the research team through stakeholders workshops. Linda and Terry will lead accessibility reviews of all materials. We will hold sessions with people from high-deprivation areas and ethnic minority background.
WS 2 & 3 - Testing and Trial: Panel members will help shape and advise during field testing and trial, providing real-time feedback on critical trial issues such as recruitment and retention. They'll shape interview guides, and co-author plain English summaries of findings.
WS4 - Implementation: PPIE partners will input into our implementation toolkit and test results in real NHS settings, ensuring it works for diverse users and contexts.
WS5 - Dissemination: Linda and Terry will co-develop accessible outputs including Easy Read summaries and subtitled videos, plus lead public engagement through meetings and webinars and community networks.
For detailed PPIE activities by WS, see the Detailed Research Plan sections on WS1-5.

Governance and Support
PPI co-investigators hold full membership of the Programme Management Group. They will be given comprehensive support for their role including training, equipment loans, and flexible meeting arrangements. All contributions are fairly remunerated according to NIHR payment guidance.

Measuring Impact
We will monitor and reflect on impact through GRIPP2 logs and review involvement activity at each milestone to ensure it is meaningful and valuable.

Investment in Partnership
Our £30,331 PPIE budget reflects genuine commitment: £15,160 for contributor payments and access costs (based on NIHR's 2025 rates), plus £15,171 for the PPIE lead's coordination and engagement support.

SUMMARY BUDGET

Justification of Costs
We request £3.42m in research costs to deliver a definitive evaluation of TACT+, a hybrid clinical-digital pathway for hearing and cognition. Costs have been benchmarked against NIHR guidance and structured around deliverables. All research (non-NHS) costs are included in this request.

NIHR Programme Grants for Applied Research Programme

1. Lead & Co-applicants (£994.4k)
Fourteen co-applicants contribute clinical, digital, methodological and PPI multidisciplinary expertise across all workstreams. FTEs were increased from Stage 1 in response to panel concern over low time allocations. All costs follow institutional salary rates and match peak activity periods. An overall Programme Manager is necessary to manage and integration across work streams, disciplines and sites. CCTU co-applicants (statistical, health economic and governance leads) are also included here. Their roles are essential to ensure delivery pace, trial oversight, and rigour in a multi-site hybrid RCT. Their inclusion follows NIHR guidance on embedding experienced trialists in complex evaluations.

2. CCTU Delivery Staff (£930.5k)
The UCL CCTU provides dedicated delivery support including:
- 1.0 FTE Trial Manager
- 0.5 FTE Clinical Project Manager
- 0.5 FTE Statistician
- 0.3 FTE Health Economist
- 0.3 FTE Data Scientist
- 0.5 FTE Programme Administrator
- 0.7 FTE Data Manager
- 0.2 FTE Database Programmer
- 0.3 FTE Site Set-up Manager

These roles deliver trial set-up, data management, statistical analysis, adherence tracking, and regulatory reporting. CCTU's involvement ensures MHRA, UK-GDPR and NIHR compliance and reflects UCL norms for hosted NIHR trials. These costs represent ~27% of the total and are proportionate to the trial's scale, complexity, digital components, and multi-workstream structure of TACT+.

3. Research Delivery & Capacity-Building Staff (£645.0k)
Includes:
- 1.0 FTE Research Audiologist (M7–45)
- 1.0 FTE Research Assistant (M4–45)
- 1.0 FTE Research Associate (M22–63)
- 2.0 FTE PhD fees and stipends (UCL & Nottingham)

These posts support trial delivery (field testing, onboarding, follow-up) while building UK research capacity. All include protected time and formal training plans. The total meets the NIHR PGfAR target of 10–20% capacity spend (~11% net), and directly addresses national gaps in digital audiology, behavioural trials and dementia prevention.

4. Digital Infrastructure & Development
Essential for delivering a digitally enabled NHS-ready model and ensuring reproducibility, adherence monitoring and coach communication.

App Development & Hosting (£223.5k): Quote from Glitchers, trusted ENHANCE contractor. Includes specification, build, MVP, optimisation, and NHS login. Phased, fixed-cost stages ensure deliverability and open-source release. Includes Cloud services (£34k): Secure hosting, video, secure messaging and alerts (36 months).
Database & Randomisation (£46.0k): OpenClinica SaaS and compliant eTMF system.
GDPR compliant secure research data hosting, generation and transcription costs (11.8K)

5. SOECAT Research Costs (130.5k)
Activities include time for personnel for recruitment and research data collection, EHR search, DocMail and SMS follow-up, hearing coach training, research data collection and data entry, and site coordination activities. SoECAT was validated jointly with SoECAT specialist. In addition to Research cost: NHS Service Support: £10.8K. Excess Treatment Costs: £88.6K.

5. Equipment & Consumables for intervention and trial delivery

Tympa kits x18 (£68.6k): Combined otoscopy, audiometry and wax removal with included Tympa Academy training. Enables safe, rapid assessment by trained coaches in community hubs.
Self-fit hearing aids x291 (£145.5k): Based on the average retail cost of the OTC models under consideration for TACT+, plus a 20% contingency to account for loss, damage, or early technical failure. We are in discussion with the suppliers of current OTC UK models, all of whom have indicated potential preferential pricing. The figure reflects prudent forward planning while remaining cost-conscious.
Digital and Research Inclusion: 278 tablet devices at £110 each for participants lacking digital access (£30.6K) and data connectivity - SIM cards (£68K). Aligned with NIHR inclusion guidance.
Laptop x7 (£5.2K)
Encrypted digital recorders (£0.6K)
Programme postage and printing costs: (£6K)
Attendance catering and promotion costs (£1K)
CCTU recruitment costs (£1.9K)
Patient folder, ISF and TMF (£5.1K)
Participant vouchers to compensate for time (£34.8K)

6. Trial and Dissemination Travel and Subsistence
Programme Meetings (£17K): TSC, TMG, CTU, IDMC, Site, for cause and routine monitoring meetings.
Participant travel fee (£18.5K): to support transport and caring costs for participants or home visits by trial delivery team if required.
UK and International conference for dissemination (£5.5K)

7 Working with People and communities
PPI related fee - participation, travel, workshop, organisation and coordination (£30.3K)

NIHR Programme Grants for Applied Research Programme

ACKNOWLEDGEMENT AND CONFLICTS

I have read and understood the terms of applying as Lead Applicant and accept this role	✓

Potential Conflicts	The applicants declare no known conflicts of interest in relation to this application. Should any potential conflict arise, it will be disclosed to the funder in accordance with NIHR policies. The ENHANCE app, to be adapted within this programme, was developed through a previous NIHR-funded project (NIHR203670). It is a non-commercial, publicly funded tool with no financial interests held by the applicants; North London NHS Foundation Trust holds the Intellectual Property (IP) and options for future potential commercialisation rights for the ENHANCE app, and will also hold these rights for the TACT+ version when developed. The inclusion of the ENHANCE app reflects research continuity and stakeholder co-design, not a financial or proprietary interest held by the applicants. Dr Devina Maru is a Clinical Advisor on the Strategy Team for the NIHR-UCLH BRC Deafness and Hearing Problems Theme, a member of the NICE Quality Standards Advisory Committee, and a Clinical Advisor to the RCGP. These roles are unrelated to this application and do not represent a conflict of interest.

Lead Applicant:	Sergi Costafreda Gonzalez	**Application Number:**	NIHR501279
Contracting Organisation:	NORTH LONDON NHS FOUNDATION TRUST		
Proposed Start Date:	01/04/2026	**End Date:**	30/09/2031

	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Total Cost to NIHR
STAFF COSTS							
Lead and co-applicants	£178,454.66	£182,267.20	£188,583.75	£191,153.40	£192,801.93	£61,138.97	**£994,399.91**
Research staff	£0.00	£83,328.88	£113,429.63	£116,489.86	£119,690.85	£30,194.11	**£463,133.33**
Other staff	£21,492.50	£88,983.98	£139,616.29	£95,098.57	£97,571.13	£24,600.98	**£467,363.45**
Total staff costs	**£199,947.16**	**£354,580.06**	**£441,629.67**	**£402,741.83**	**£410,063.91**	**£115,934.06**	**£1,924,896.69**
OTHER RESEARCH/ AWARD COSTS							
Travel and subsistence	£3,355.67	£6,574.13	£7,615.27	£7,615.27	£7,615.27	£2,744.40	**£35,520.01**
Equipment	£0.00	£12,459.78	£24,953.70	£24,919.56	£6,246.96	£0.00	**£68,580.00**
Consumables	£30,477.23	£38,578.46	£42,199.52	£35,876.97	£32,724.36	£14,253.47	**£194,110.01**
Knowledge mobilisation, dissemination and impact	£1,090.09	£2,593.08	£2,590.10	£1,090.09	£1,090.09	£3,046.54	**£11,499.99**
Training and development	£78,669.68	£156,110.51	£195,676.59	£158,979.23	£46,967.47	£8,548.22	**£644,951.70**
Research inclusion	£8,815.43	£28,135.62	£35,256.53	£17,596.99	£8,815.43	£0.00	**£98,620.00**
Working with people and communities	£10,629.85	£6,769.77	£2,367.95	£2,695.80	£5,303.88	£2,563.97	**£30,331.22**
Other direct costs	£44,905.46	£77,446.01	£99,525.43	£87,703.91	£79,035.82	£23,228.21	**£411,844.84**
HEI indirect costs	£0.00	£0.00	£0.00	£0.00	£0.00	£0.00	**£0.00**
Other partner organisation indirect costs	£0.00	£0.00	£0.00	£0.00	£0.00	£0.00	**£0.00**
TOTAL COST TO NIHR	**£377,890.57**	**£683,247.42**	**£851,814.76**	**£739,219.65**	**£597,863.19**	**£170,318.87**	**£3,420,354.46**
Co-funding from other sources							**£0.00**
Cost to NIHR							**£3,420,354.46**
NHS support costs	£1,803.83	£1,803.83	£1,803.83	£1,803.83	£1,803.83	£1,803.83	**£10,822.98**
NHS excess treatment costs	£14,765.68	£14,765.68	£14,765.68	£14,765.68	£14,765.68	£14,765.68	**£88,594.08**
Non-NHS excess treatment costs	£0.00	£0.00	£0.00	£0.00	£0.00	£0.00	**£0.00**
TOTAL FUNDING							

Application Budget: NIHR501279

REQUESTED	£394,460.08	£699,816.93	£868,384.27	£755,789.16	£614,432.70	£186,888.38	**£3,519,771.52**

Detailed Research Plan Template for Domestic Programmes and Personal Awards

****Background and Rationale****

1. Converging Challenges: Cognitive Decline, Hearing And Health Inequality

Dementia is a growing public health crisis, affecting over 55 million people worldwide, with UK cases set to exceed 1 million by 2025 [1,2]. Up to 40% of dementia risk is attributable to modifiable risk factors, making prevention both an opportunity and a priority [3]. Age-adjusted dementia rates have declined in high-income countries and affluent groups, likely due to better cardiovascular health, but risk has increased in under-served communities [3].

Age-related hearing loss is a key, modifiable risk factor; every 10 dB of loss is associated with 16% increased dementia risk [3,4]. It affects over a quarter of UK adults aged 50-89, and is more common among people with lower income or education [5]. Untreated hearing loss harms communication, mood, social life, and quality of life, and strongly raises dementia risk [3,6].

The solution seems simple: hearing aids. Yet, fewer than one in seven older adults who could benefit use them. This gap is widest in under-served groups, who are both more likely to have hearing loss and less likely to get help and use hearing aids [5,7,8]. Lower socioeconomic status, limited education, and living alone all worsen the gap. Bridging it with scalable, inclusive interventions is a major chance to reduce dementia risk and tackle health inequalities.

The urgency of addressing this gap has recently been underscored by the 2025 James Lind Alliance Priority Setting Partnership on co-existing dementia and hearing loss [9] This national consultation brought together patients, carers, and professionals to identify the top research priorities. These included whether managing hearing loss can reduce dementia risk, how to integrate cognitive and hearing assessments, and how best to deliver joined-up care. TACT+ responds directly to these priorities by targeting an underserved group at high risk of dementia with a scalable hearing intervention embedded in primary care.

2. The Target Population: Mild Cognitive Impairment

Our target population is people with Mild Cognitive Impairment (MCI), the stage between healthy ageing and dementia. MCI affects around 6% of adults aged ≥ 60, rising to about 25% by age 80 [10,11]. It involves cognitive decline without major functional loss, but carries a high risk of progressing to dementia, between 14% and 23% within 18 months [12]. These individuals often fall into a diagnostic and therapeutic gap in the NHS. Most are managed in primary care, where cognitive difficulties can make care pathways, including hearing care, hard to navigate [13]. Those seen in memory clinics are often discharged without a structured plan [14,15]

People with MCI are ideal for prevention: they are at high risk, able to use hearing aids [16,17], and can engage with a programme like TACT+. Our pilot study [18], showed that hearing aid fitting plus non-specialist coaching in Memory Services led to 75% hearing aid use, versus 22% in standard care, a 53% difference [95%CI:

29%–77%] [18]. Though not powered for efficacy, this study also showed promising signals on cognition, quality of life, and mood.

Figure 1. Examples of Self-Fitting Hearing Aids Under Consideration for TACT+ Trial

3. The TACT+ Solution: A Primary Care Pathway For Supported Self-Fitting

The recent UK policy enabling over-the-counter (OTC), self-fitting hearing aids, creates a major opportunity for scalable, low-cost intervention [19]. These devices let users adjust settings to their needs and are often styled like consumer earphones, helping normalise use and reducing stigma. Randomised trials show that for mild-to-moderate uncomplicated hearing loss, they can deliver outcomes similar to standard audiology-fitted aids—along with the wider communication and psychosocial benefits of better hearing [20].

But a key question remains: can older adults with early cognitive decline adopt hearing aids with structured but non-specialist support?

This programme addresses that question and the main barrier to scale: whether *supported* self-fitting can unlock hearing aids' potential [20,21,22] for cognitive preservation in older people with MCI.

We propose TACT+ (Treating Auditory impairment to preserve Cognition) as a new pathway, embedded in primary care networks, combining assessment and supported self-fitting for older adults with cognitive difficulties. Research will be base in primary care hubs accepting referrals from local practices and memory clinics. Cognition will be screened and hearing tested using handheld Tympa devices by nurses. The model blends accessible self-fitting aids with light-touch remote coaching to address the biopsychosocial and behavioural elements of using hearing aids. This support builds confidence and sustained use—empowering people, especially underserved groups, to protect their hearing and cognitive health.

4. The Evidence for Intervention

While the implementation gap is clear, a key clinical question remains for dementia prevention: does treating hearing loss slow cognitive decline in people with MCI? The current evidence is promising but not yet definitive for NHS policy.

The case for the TACT+ trial is based on three converging RCT evidence streams:

- First, modern self-fitting hearing aids achieve outcomes similar to audiologist-fitted aids for uncomplicated mild-to-moderate hearing loss [21,22]. This supports self-fitting as a sound, scalable intervention model.

- Second, RCTs show that OTC self-fitted hearing aids improve not just hearing, but also broader psychosocial outcomes, critical drivers of well-being in older adults [20].
- Third, and most relevant here, there is a strong signal from ACHIEVE trial - the first RCT of hearing aids in older adults - that hearing interventions can slow cognitive decline in the those most at-risk of dementia, similar to MCI populations [16,17].

The key studies supporting these findings are summarised in Table 1.

Table 1. Summary of key RCT evidence underpinning TACT+

Study / Evidence Source	Population	Key Finding(s)
ACHIEVE Trial [16,17,23]	977 older adults (70-84 yrs) with hearing loss.	No overall effect, but in high-risk group (low baseline cognition): 62% less cognitive decline. Validates targeting MCI for prevention.
TACT Pilot Trial [18]	58 adults with MCI in NHS Memory Services.	Hearing aid use: 22% → 75%. Signals of improvement in ACE-III and memory, quality of life, and mood. Shows feasibility and impact of supported model in MCI.
De Sousa et al. [21,22]	64 adults with mild-to-moderate hearing loss.	OTC self-fitting ~= audiologist-fit on audiological outcomes at 8 months. Confirms self-fitting model viable.
Jiang et al. [20]	221 older adults with hearing loss.	OTC + healthcare worker support improved depression, isolation, and communication. Shows wider value of supported self-fitting.

5. The Plausible Mechanisms of Action

The strong signals from recent trials [16,18] are supported by clear biological, psychosocial and behavioural mechanisms. A recent review and theoretical synthesis concluded that treating hearing loss may delay, rather than prevent, dementia progression [24].

Treating hearing loss may protect cognitive function through three interrelated pathways:

1. **Reducing cognitive load**: Hearing loss increases mental effort needed to follow speech, drawing resources away from memory and executive functions and accelerating decline [25,26]. Our PPIE highlighted this: "When you've got hearing

loss, you come back from a whole day of listening to people and you're incredibly tired."; "With hearing aids, I can concentrate better."

2. **Mitigating neurodegeneration**: Each 10 dB loss raises dementia risk by 16% [4]. Imaging shows faster brain shrinkage in auditory and cognitive areas in people with untreated hearing loss, along with higher tau accumulation [27]
3. **Reversing low mood and social isolation**: Hearing loss often leads to gradual social withdrawal, worsening mood and shrinking social networks—both dementia risk factors [3]. PPIE members emphasised the broader social and psychological impact: "*There's still a lot of prejudice about hearing aids....*"; social reconnection can be enabled by hearing aids especially with coaching to overcome fear of stigma.

6. The Need for a Definitive Trial

Current evidence strongly suggests that supported self-fitting of hearing aids is a viable and beneficial intervention. But a strong signal is not proof. Before this pathway can be recommended for national adoption, a large-scale randomised controlled trial (RCT) is needed to answer three key questions:

1. **Is there a clinically meaningful cognitive benefit?**
   We must determine if the cognitive effect seen in the ACHIEVE high-risk subgroup holds in an NHS-based MCI population [16, 17]. This is a major public health question. Our TACT pilot found that the potential to protect memory was a key motivator for patients—even those unaware of hearing loss—to use hearing aids [18]. A clear answer is needed to guide clinical practice and public messaging.
2. **What are the broader patient and partner benefits?**
   Hearing loss is not just a dementia risk factor; it harms communication, social life, and quality of life [28]. This trial will measure the full impact of TACT+ on these outcomes. Our pilot suggested gains in mood and social participation, but only a larger trial can confirm their size and importance for patients and their communication partners. These psychosocial outcomes matter in their own right, regardless of cognitive effects.
3. **Is it cost-effective for the NHS?**
   Scaling up hearing care in primary care requires investment. We must show that the TACT+ model is not only effective but cost-efficient. Modelling suggests hearing aids could be cost-saving if they delay dementia onset [29], but this must be tested with trial data.

Only a definitive Programme Grant—combining RCT, economic and process evaluation, and co-design by clinicians, methodologists, and people with lived experience—can deliver the robust evidence the NHS needs to act.

7. National Strategic Alignment

TACT+ supports key NHS and NIHR priorities [30], delivering three NHS 10-Year Plan shifts: care closer to home (via Primary Care Networks and Neighbourhood Health Centres), digital innovation (hearing assessment, OTC aids, remote coaching), and prevention (targeting MCI before dementia).

By tackling hearing loss in underserved groups—a major modifiable dementia risk—TACT+ directly addresses DHSC's Areas of Research Interest, NIHR's Themed Call on Prevention (which includes hearing loss), and the Highlight Notice on Dementia, which emphasises scalable implementation [31]. It also supports the GIRFT programme's aims for standardised, high-value NHS care [32].

****Aims and objectives****

Aim: To co-develop, evaluate, and prepare for national implementation of TACT+: a supported self-fitting hearing aid pathway for people with MCI, through co-production, feasibility testing, a definitive trial, implementation research, and dissemination.

Table 2. SMART objectives and mapping to workstreams:

#	Objective	Key success criteria	Work-stream
1	Co-produce the TACT+ pathway (hearing protocol, self-fitting model, coaching, app) with patients, carers, clinicians, and commissioners	Finalised intervention and materials signed off; EDI embedded; ≥80% stakeholder agreement.	WS1 Months 1-12
2	Field-test the pathway in two NHS sites (London, Manchester) to assess acceptability, feasibility and operations.	Recruitment achieved and diverse; ≥80% successful self-fitting at 4 weeks; App System usability score ≥70.	WS2 Months 7-18
3	Assess clinical and cost-effectiveness in a multicentre RCT (primary outcome: MoCA at 18 months).	Trial powered to detect ≥1.5 MoCA points; economic evaluation completed.	WS3 Months 16-63
4	Develop an implementation strategy and Toolkit using a process evaluation to identify barriers/enablers.	Components mapped to NPT/TDF; Toolkit tested in 2 hubs with ≥80% fidelity and acceptability.	WS4 Months 1-63
5	Maximise impact by disseminating findings and supporting Toolkit uptake, focusing on under-served groups.	Toolkit outputs delivered (e.g., e-learning, budget tool); engagement with NHS/NICE.	WS5 Months 36-63
6	Finalise NIHR Report	All other activity concluded; CI finalises report and responds to queries.	Months 63-66

Note. MoCA, Montreal Cognitive Assessment; NPT, Normalisation Process Theory; RCT, randomised controlled trial; TDF, Theoretical Domains Framework.

****Methodology / Plan****

WS1: Co-design
Months 1-12
Milestone: Locked intervention and blueprint for WS2

This workstream will co-produce a trial-ready version of the TACT+ intervention. Building on our secondary care pilot [18], we will adapt it for primary care-led delivery

using self-fitting hearing aids and remote support. The goal is to create a model that is acceptable, equitable, and feasible for NHS implementation.

1. Co-Production and Behavioural Design

The TACT+ model will be co-designed with people with lived experience embedded as equal contributors. We will apply the Behaviour Change Wheel, COM-B model, and Theoretical Domains Framework (TDF) [33] to identify behavioural barriers and enablers among patients (e.g. motivation, confidence, opportunity) and staff (e.g. capability to deliver assessment and coaching). These insights will guide design of referral, assessment, delivery, and training processes.

2. Co-Design Workshops (Months 5–9)

Mixed-stakeholder workshops will co-develop:

- **Referral Pathway:** To maximise reach while maintaining feasibility, we anticipate using hub-and-spoke model: referrals from local GP practices and memory services ("spokes") into a small number of PCN-based assessment hubs. Recruiting from primary care requires wide geographical coverage, but placing a Tympa-trained clinician in every GP practice is unworkable. Centralising delivery in hubs achieves a balance between accessibility and efficiency, with hubs located near patients' usual services. This "primary care plus" model has been used in other areas such as MSK rehabilitation, and was piloted in audiology using Tympa technology and remote supervision [34]. One of our co-investigators, Prof Nishchay Mehta, co-authored that work and will help adapt the model for TACT+. This approach is also well aligned with the NHS 10-Year Plan [30] and the development of Neighbourhood Health Centres, which promote integrated services delivered closer to home.
- **Consent and Randomisation:** Participant-friendly explanations of the research and randomisation process.
- **Assessment Protocol:** We anticipate to use the integrated Tympa System technology (ear exam, wax removal, audiometry) [34,35] which can be delivered by trained nurse, with research audiologist oversight and protocolised communication and referral strategies for red-flag findings.
- **Cognitive and Psychosocial Measures:** Montreal Cognitive Assessment (MoCA) will be used both as a screening tool and as the primary outcome. MoCA alone cannot diagnose MCI, so results will be framed as identifying those who may need extra support with hearing care. Those with memory concerns will be advised to consult their GP.
- **Device Selection:** Criteria include suitability for mild-to-moderate hearing loss, ease of self-fit and use and low maintenance (e.g. rechargeable). User and clinician feedback will shape final choices. Products under review include devices from Sennheiser, Jabra and Apple (Fig. 1), with more could be reviewed as this is a fast-moving area between now and selection. We will only select devices that, at the point of procurement, hold a valid UKCA or CE mark for self-fit hearing-aid use and will be deployed within their intended use, therefore not requiring MHRA approval.

- **Hearing Coach Support:** Building on the TACT pilot roles [18], trained coaches will use behaviour change techniques (e.g. SMART goals [36] and the GROW model (Goal, Reality, Options, Will [37]) to support hearing aid use and confidence. Anticipated activities will include:
    - Setting and tracking personalised hearing goals (via Client Oriented Scale of Improvement - COSI [38])
    - Explaining the link between hearing and cognition and individual benefits
    - Problem solving behavioural/social/emotional worries
    - Troubleshooting barriers to use
    - Coaches will be supervised weekly and supported clinically by a research audiologist. Hearing aid usage will be tracked (where feasible) to allow tailored responses.
- **Digital Support via the TACT+ App**: The TACT+ app will build on the ENHANCE platform [39] (Fig. 2), a secure digital environment for cognitive training and health coaching. During WS1, content will be re-skinned for hearing-rehabilitation tasks and adapted for memory-friendly navigation. Design adaptations will include:
    - Simplified user interface, step-by-step captioned video guides, and easy access to coach support via messaging and video calling. This ensures alignment with the needs of older adults with MCI and supports sustained use of hearing aids in daily life.
    - Instructional videos on hearing and communication
    - Auditory rehabilitation games
    - Step-by-step videos on hearing aid use
    - Goal tracking and reminders, motivational “meadow” reward interface
    - Easy access to coach support via messaging and video calling
- Communication partners (e.g. spouses or friends) will be optionally involved to support hearing aid use, if the participant consents.

Figure 2. The “Meadow” interface in the ENHANCE app, shown here with the blood pressure module. For TACT+, this will be adapted to support hearing-related activities and coach interaction.

3. Usability Testing and Inclusion

The app will be tested with 12 representative users using think-aloud protocols. We aim for a System Usability Scale (SUS) score ≥70 [40]. Iterative feedback will ensure accessibility for people with MCI and low digital literacy. The app will use large font, simple language, and pictorial cues to ensure usability by people with low health or digital literacy. Participants without digital access will be loaned 4G-enabled tablets; coaches will offer phone-based support as needed.

4. Finalisation and Protocol Development (Months 10–12)

We will finalise the intervention and trial protocol through multidisciplinary working groups. Outputs will include:

- SOPs and manuals (recruitment, assessment, coaching, app use).
- Accessible, co-designed participant materials.
- Updated logic model and fidelity checklist.
- Modular training curricula for staff and coaches (Table 2).

Table 3. Anticipated Roles and Supervision Structure for TACT+ Intervention

Role	Core duties	Supervision
Nurse assessor (band 4/5 nurses or HCA)	Deliver baseline/follow-up at hubs (otoscopy, Tympa test, MoCA, psychosocial tools); flag red-flag findings.	Case reviews with the Research Audiologist.
Hearing coach (band 4)	Support self-fitting and app-based behaviour change; track usage; set goals; troubleshoot basic device or app issues.	Weekly group supervision + ad hoc escalation to Audiologist.
Research Audiologist (HCPC-registered)	Clinical lead; validate red-flags; adjust hearing aid parameters; oversee coach training and fidelity.	Reports to CI; attends monthly trial meetings.

WS2: Field testing and optimisation
Months 7-18
Milestone: Refined TACT+ package ready for WS3

1. Introduction and Aims of WS2

WS2 will assess the real-world feasibility of the TACT+ intervention co-developed in WS1. We will implement and optimise a scalable, primary care–based model that integrates supported self-fitting of hearing aids with behavioural coaching. This workstream will de-risk the full trial by:

- Testing the clinical and operational feasibility of delivering TACT+ in PCN hubs, using the triad of delivery roles: Nurse Assessor, hearing coach, and supervising Research Audiologist.
- Optimising safety, usability and acceptability by identifying and addressing workflow challenges, coaching queries, and clinical red-flag management issues.
- Producing trial-ready SOPs, training manuals, and a refined version of the TACT+ app and coaching protocol for WS3.

2. Settings and Participant Recruitment

We will field-test the TACT+ intervention in two demographically and structurally distinct metropolitan areas: North London and Greater Manchester. Each will host a

Primary Care Network (PCN) hub delivering all research activities—from consent to baseline assessment and hearing aid fitting— in a standardised, “one-stop” model.

Participants will be recruited via two pathways:

Pathway	Source	Value Added
Primary-care: list search + opportunistic case-finding	GP records + mail-outs to adults ≥55 in IMD deciles 1-4	Identifies older adults with possible cognitive frailty managed in primary care
Memory-clinic Patient-Identification-Centre (PIC)	Recent MCI diagnoses from NHS Memory Services	Adds minority of participants with formally diagnosed MCI

Primary care recruitment will focus on underserved groups, targeting high-deprivation postcodes. Materials will be designed for accessibility, including easy-read English. Recruitment will last 3 months, with a 4-week post-ran.

We will screen GP records to identify people likely to be eligible. Personalised invitations will be sent to those aged ≥55, living in the community (not in residential or nursing care), with no recorded diagnosis of dementia, progressive neurological disease (other than MCI), severe mental illness, or active substance misuse.

Carer involvement was limited in our pilot [18], likely because participants were functionally independent and family or friends did not identify as “carers.” In this project, and based on PPI input, we will explore involving a nominated “communication partner” (e.g. spouse or friend) who, with participant consent, can receive updates from the coach and support hearing aid use.

3. Field testing sample-size justification

We will recruit 62 participants (31 per site) to estimate key feasibility outcomes with adequate precision for main trial planning [41]. For an expected retention rate of 80%, this sample yields a 95% confidence interval of ±9.9%. It also gives >90% probability of detecting at least one operational issue or adverse event with a true incidence of 4% or more, thus providing high confidence in identifying major risks before progressing to the full RCT.

4. Standardised Assessment and Randomisation Pathway

All WS2 research procedures will take place in a Primary Care Network (PCN) hub, located within a GP practice or Neighbourhood Health Centre. The full participant journey—consent, baseline assessment, hearing test, randomisation, and first coach meeting to support hearing aid self-fitting —will be completed at this single site to support consistency, scalability, and efficient delivery.

4.1 Nurse Assessor: Training and Scope

Each hub will be staffed by a trained Nurse Assessor, responsible for all baseline and follow-up assessments. They will complete TympaHealth Academy training in

digital otoscopy, microsuction, and pure-tone audiometry, and be remotely supervised by a Research Audiologist for eligibility and referrals. All procedures will be logged, exclusions reviewed, and 10% of ear-canal images audited by a remote ENT specialist. This model is supported by evidence that non-specialist staff can safely and effectively deliver digital hearing assessments in primary care [34,35].

4.2 Baseline visit workflow

1. Consent and Screening

Consent will be obtained by the Nurse Assessor prior to any assessments. Participants must have capacity to consent.

2. Ear and Audiological Assessment

The Nurse Assessor will perform a protocolised hearing assessment using Tympa:

- **Current hearing aid use**: Assessed first using item 1 of the *International Outcome Inventory for Hearing Aids* (IOI-HA) [42]. Participants currently using hearing aids will be excluded from further assessment.
- **Digital otoscopy**: the ear canal and tympanic membrane are visualised and imaged. Occluding cerumen will be removed via microsuction if needed.
- **Red-flag checklist:** A standardised tool screens for otological red flags (e.g. pain, discharge, dizziness, sudden/asymmetrical hearing loss), in line with BAA guidance [43].
- **Audiometry:** Pure-tone audiometry (0.5 - 5 kHz) using Tympa to determine hearing thresholds in each ear.

Participants may proceed if they are not using hearing aids, no red flags are found and audiometry confirms symmetrical, mild-to-moderate sensorineural loss (30–60 dB HL in the better ear). Nurse Assessors document all findings for asynchronous review by Research Audiologist. Any red flags or uncertainty will be discussed. This model ensures oversight while supporting scalability.

3. Outcome Assessments

Once audiological eligibility is confirmed, the Nurse Assessor will administer a standardised set of outcome measures. Two of these (MoCA and C-IADL) also serve as eligibility screens, with thresholds detailed in Section 4.3.

- **Primary Cognitive Outcome**: The Montreal Cognitive Assessment (MoCA) will confirm eligibility (score 18–25, with +1 for ≤12 years of education) and serve as the trial's primary cognitive outcome. It is sensitive to early decline and suitable for repeated use [44].
- **Functional Status**: The Lawton-Broday Instrumental Activities of Daily Living (IADL) assesses functional independence across eight domains using a binary scale (independent = 1; needs help = 0) [45], with adjustments to reduce gender and cultural bias [46].
- **Perceived Hearing Difficulty**: The Hearing Handicap Inventory for the Elderly – Screening, HHIE-S [47] to assess communication difficulties related to hearing loss.

- **Psychosocial Wellbeing:** The Geriatric Depression Scale - short form (GDS-15) [48] and Lubben Social Network Scale (LSNS-6) [49] to assess mood and social connections.
- **Health Economics:** The Client Service Receipt Inventory (CSRI) [50] and Health Utilities Index Mark 3 (HUI3) [51,52] will record health service use and health-related quality of life.

All measures will be repeated post-intervention to assess 4-week change.

4.3 Eligibility summary and randomisation. Participants must meet all the following criteria to be randomised:

General Criteria:

- Aged ≥55 years
- Living in the community (not in residential or nursing care)
- No recorded diagnosis of dementia or other progressive neurological disorder (excluding MCI)
- No severe mental illness or active substance misuse

Cognitive Criteria (operationalised MCI Definition):

- MoCA score 18–25 inclusive (+1 point added for ≤12 years of education) [44]
- Preserved functional independence: score ≥7/8 on the IADL, which reliably excludes dementia [45,46]

Hearing Criteria:

- Not currently using hearing aids (confirmed via item 1 of the IOI-HA [42]).
- Symmetrical, mild-to-moderate sensorineural hearing loss (30–60 dB HL in better ear), confirmed by Tympa audiometry
- No red-flag otological signs or unresolved occluding wax [43]
- No. If a MoCA score <25 is unexpected (e.g. in primary care recruits), participants will receive personalised feedback and a letter suggesting they speak to their GP if concerned about memory. Similarly, participants will receive supportive explanations if hearing loss is newly identified. We will adapt the script used successfully in our pilot, to explain results and encourage engagement with hearing support.

Randomisation will occur after all baseline data are complete, using a secure web-based system with 2:1 allocation (TACT+ : control), to prioritise learning from the intervention arm.

5. The TACT+ Active Intervention

Participants randomised to the intervention arm will receive a structured, multi-component hearing support programme comprising: immediate self-fitting of hearing aids with guidance, personalised coaching, and digital app-based reinforcement.

5.1. Intervention onboarding: A flexible 'one-stop' approach.

All intervention activities take place at the PCN hub. Based on pilot learning, we will offer two onboarding formats to accommodate participant preferences and fatigue:

- **Default pathway (single visit)**: Immediately after randomisation, participants will self-fit their hearing aids with support from the coach, followed by technology setup and initial coaching onboarding.
- **Split-visit pathway**: For participants who are tired or prefer more time, a second visit will be scheduled to complete hearing aid fitting and coaching onboarding.

We will assess uptake and impact on engagement and retention of each pathway.

5.2 Core components of the TACT+ intervention

A. Self-fitting hearing aids

Participants will self-fit hearing aids selected during WS1, supported by their Hearing Coach. Devices will be chosen based on usability, suitability for mild-to-moderate loss, and compatibility with self-fitting protocols. Where feasible, audiometric thresholds from the Tympa system will be used to generate an initial, evidence-based fitting based on a validated prescriptive formula (e.g., NAL-NL2), configured using the manufacturer's proprietary software or in-app calibration tools.

The hearing coach will support physical fit, comfort, and use of controls. A brief motivational conversation will link hearing to cognition, mood, and social participation—an approach shown to enhance engagement in our pilot .

B. Personalised hearing coaching:

Each participant is supported by a dedicated hearing coach (psychology assistant or other health worker) trained in behaviour change and practical hearing aid support.

Coaching Delivery (Field-Test Version):
Participants will receive a shortened version of the full coaching programme. It begins with a face-to-face onboarding session at the PCN hub, followed by remote check-ins at Weeks 1, 2, and 4 via the TACT+ app or phone. This 4-week model targets the early fitting phase, identified in our pilot as key to long-term use .

Onboarding Session Includes:

- A motivational discussion linking hearing, cognition, and quality of life
- Supervised self-fitting of the hearing aids
- Setup and orientation for the TACT+ digital support app

Follow-Up Check-ins (Weeks 1 & 4):

- Hearing aid use and troubleshooting practical and technical problems
- Setting up and checking on personalised hearing goals (using COSI [38])
- Reinforcement strategies to support habit formation and confidence

Coaches will receive weekly group supervision from the Research Audiologist and escalate clinical or technical issues as needed.

C. The TACT+ digital support app

Coaching is delivered via the TACT+ app, adapted from the NIHR-funded ENHANCE platform [39], which is co-designed for older adults and owned by NLFT (the sponsor of both ENHANCE and TACT+). It will support scalable delivery for people with MCI.

During onboarding, the coach ensures participants can confidently use the app, either on their own device or a study-issued 4G-enabled tablet.

Core Features:

- **Instructional support**: Short videos on hearing, how to optimise communication, and pictorial manuals for setup, cleaning, and troubleshooting of hearing aids
- **Engagement tools**: Reminders and auditory rehabilitation games to build habit and confidence
- **Coach access**: Secure messaging and video calls
- **Motivational design**: A "meadow" interface that rewards participation (e.g. flowers appear as tasks are completed)

Participants without a suitable device will get loan tablets with 4G data. For low digital literacy we will print guides and telephone-based support available.

D. Follow-up and post-trial care
At 4-week follow-up, participants will complete the full baseline outcome battery, including the IOI-HA to assess use and benefit. Referral to NHS audiology will be offered only on request, if hearing aid use is not established, or if recommended by the Research Audiologist.

6. The Control Arm: Standard Care

Participants in the control arm will receive care aligned with current NHS pathways. After baseline assessment at the PCN hub, they will be given a printed summary of their hearing results with a letter encouraging discussion with their GP, and potential referral to NHS audiology or private providers. If NHS self-referral to audiology is available during the field test (as outlined in the NHS 10-Year Plan[30]), this option will also be offered to ensure alignment with evolving policy and equitable access.

At the 4-week follow-up, we will record any hearing-related actions taken by control participants, including contact with their GP, local audiology services, or independently accessed care, as well as any hearing aid prescription or use.

7. Analysis of Field Testing Data

7.1 Feasibility and Progression Metrics (Quantitative)

We will assess trial readiness through predefined metrics:

- **Recruitment and retention:** consented, randomised, followed-up at 4 weeks
- **Adherence and engagement**: The main endpoint is hearing aid use at 4 weeks, a key acclimatisation window linked to long-term use. We will measure:
    - Objective hearing aid use via device logs
    - Self-reported daily use
    - Engagement with the TACT+ app (logins, videos, goal tracking)
- **Equity of reach**: Participant IMD quintile and self-reported ethnicity will be monitored to assess reach into underserved populations.
- **Pathway acceptability**: Uptake of split- vs single-visit onboarding will be recorded to inform future delivery.

Table 4. Progression criteria

	Red (Stop)	Amber (Amend)	Green (Go)
Total randomised (by 3 months)	<30	30-61	>=62
TACT+ delivery (fitted by 4 weeks)	<50 %	50-79 %	≥ 80 %
Retention at 4 weeks follow-up	<50 %	50-89%	≥ 90 %

7.2 Acceptability and experience (qualitative)

We will conduct qualitative interviews with ~25 participants (n=15 intervention, n=10 control) and professionals (primary care staff, nurse assessors, hearing coaches, research audiologists, n =~15). Sampling will include both successful and unsuccessful self-fitting cases.

Topics will include:

- Usability and acceptability of procedures and components (hearing aids, coaching, app)
- Barriers to engagement, including cognitive, digital, or practical challenges
- Staff experience with training, delivery, and oversight

Topic guides will draw on Normalisation Process Theory (NPT) to explore implementation challenges and inform WS4.

7.3 Fidelity monitoring

Fidelity monitoring will be undertaken during WS2 to inform iterative refinement. It will include structured coach logs and limited audio recordings or session observations (with consent). Detailed fidelity procedures and integration into process evaluation are described in WS4.

7.4 Iterative refinement and outputs

The core team, including PPI contributors, will hold regular data review meetings to refine protocols, training, and app features. All changes will be documented. By the end of WS2, we will produce Milestone 2: a finalised, field-tested intervention with optimised SOPs, training resources, and participant materials, ready for WS3.

WS3: Full RCT
Months 16-63
Milestones 3 & 4: Internal pilot completed; effectiveness, cost-effectiveness results.

1. Trial Design

Workstream 3 will deliver a multi-centre, pragmatic, two-arm randomised controlled trial (RCT) with an internal pilot. The aim is to evaluate clinical and cost-effectiveness of the TACT+ intervention versus standard care in adults with MCI and hearing loss.

The trial will implement the co-designed intervention from WS1, refined through WS2 field testing. A parallel process evaluation (see WS4) will assess fidelity, acceptability, and implementation.

A CONSORT flow diagram is provided in the Uploads section. The trial will adhere to CONSORT 2025 [53] and the CONSORT-EHEALTH extension [54] to cover key aspects of digital health trials, including user engagement metrics, intervention detail, usability testing, access pathway, context of use, and app iteration. Outcome assessors will be masked; engagement data will be captured via backend logs.

2. Setting and Population

Setting: We will recruit from 12 sites across the UK, each centred on a Primary Care Network (PCN) hub where all trial activities will be delivered. To ensure national representativeness, hubs will be distributed across at least four NIHR Regional Research Delivery Networks (RDNs) or devolved nations, covering urban, rural, and socioeconomically diverse areas. Regional clustering balances efficient coordination with broad coverage.

We are engaging RDNs and exploring local PCNs in the North West, North East and North Cumbria, East and West Midlands, East of England, South Central, South West Peninsula, and North London. We also have early partnerships in Wales.

Where feasible, local Memory Services will link to PCN hubs as Participant Identification Centres (PICs), referring people diagnosed with MCI.

Recruitment Feasibility: We will recruit 424 participants over 15 months - about 3 randomised per site per month. This is supported by WS2 targets (~5/month/site) and our pilot , which achieved 4/month from a single region. Transitioning to higher-throughput, primary care-led hubs further supports this rate.

Target Population: Eligibility criteria remain unchanged from Section 4.3 (WS2).

3. Intervention and Comparison

Intervention Arm (TACT+): Participants will receive self-fitting hearing aids supported by a structured three-month personalised coaching programme, delivered via the TACT+ app. Self-fitting follows manufacturer protocols with in-person support from a hearing coach. Coaching includes behaviour change techniques, goal-setting,

and remote troubleshooting. Follow-up sessions occur at weeks 1,2,4,8, and 12. To promote sustained use, participants also receive brief check-ins or booster sessions at months 6, 9 and 15, as needed. The intervention follows the WS2-refined protocol and is overseen by a Research Audiologist through weekly group supervision and ad hoc input.

Control Arm (Standard Care): As per WS2, Participants will receive a printed summary of their hearing results, plus written advice to consult their GP if desired. Where NHS audiology self-referral is available, this option will be explained. Control participants retain usual access to NHS or private hearing services.

4. Outcome Measures, Trial Duration & Schedule of Assessments

The trial's primary objective is to preserve cognitive function, a core outcome for patients, carers, and clinicians. Cognitive decline drives progression to dementia and strongly predicts quality of life and mortality. Outcomes will be assessed at baseline, 12 months, and 18 months by researchers blinded to allocation. The **primary outcome** is between-arm difference in MoCA score at 18 months, adjusted for baseline.

The **18-month endpoint** is a pragmatic choice: long enough to observe meaningful effects, short enough to minimise attrition and bias. This is supported by three lines of evidence:

- **Natural history of MCI**: In the VMCI-Tuscany cohort, individuals who progressed to dementia declined by 2.6 MoCA points over 24 months vs. 0.7 points in stable MCI, a 1.9-point gap. Our target effect size of 1.5 points at 18 months represents a meaningful proportion of this trajectory [55] and aligns with the magnitude of change in cognitive decline we observed through electronic health records review of our London memory clinic cohort (n=526).
- **Mechanistic plausibility**: Behavioural change—hearing aid use, reduced listening effort, increased engagement—consolidates over 6–12 months. Cognitive benefits are expected to follow.
- **Attrition mitigation**: Longer follow-up increases missing data. In VMCI-Tuscany, 27% had dropped out by 24 months. Capturing the primary endpoint at 18 months balances sensitivity to change and retention.

To support this, all secondary outcomes—including cognition—will also be collected at 12 months with additional 6-month telephone follow-up to collect health and social care resource use via CSRI. These will provide interim read-outs to counter memory biases expected to be higher in our population, will inform imputation models, and will capture earlier behavioural effects with minimal additional burden.

Table 5. Schedule of Assessments

Domain / Measure	Instrument	Baseline	12 m	18 m †
Eligibility & Covariates				
Functional independence	IADL [45,46]	✓	–	–

Hearing thresholds / red flags	Tympa otoscopy + PTA0.5-4 kHz	✓	–	–
Demographics	Age, sex, education	✓	–	–
Primary Outcome				
Global cognition	MoCA [44]	✓	✓	✓
Secondary Outcome				
Psychosocial				
Mood	GDS-15 [48]	✓	✓	✓
Social connection	LSNS-6 [49]	✓	✓	✓
Hearing Outcomes				
Perceived handicap	HHIE-S [47]	✓	✓	✓
Hearing-aid benefit	IOI-HA [42]	–	✓	✓
Hearing aid use ‡	Data-logging (h/day)	–	✓	✓
QoL & Economics				
Health utility	HUI-3 [51,52]	✓	✓	✓
Resource use*	CSRI [50]	✓	✓	✓
Clinical Progression				
Dementia diagnosis	GP / hospital records, self-report	✓	✓	✓
Fidelity & Engagement ‡				
Intervention delivery	Coach logs & TACT+ app analytics	–	✓	✓
Safety				
Adverse events	Standard AE form	✓	✓	✓

Note. † Primary endpoint; ‡ Intervention arm only; CSRI, Client Service Receipt Inventory - * additional CSRI collection by telephone at 6 months; C-IADL, Corrected Lawton-Brody IADL; GDS-15, Geriatric Depression Scale - short form; HUI-3, Health Utilities Index Mark 3; LSNS-6, Lubben Social Network Scale-6; MoCA, Montreal Cognitive Assessment.

5. Sample Size and Power

We will enrol **424 participants** to detect a between-group difference of 1.5 points on the MoCA at 18 months with **90% power** (α=0.05, two-tailed). This effect size reflects a clinically meaningful change, and is a conservative target based on longitudinal studies estimating the minimal clinically important difference (MCID) for MoCA **between 1.6–3 points** [56]**,** and endorsed by our patient and public collaborators.

We assume a **conservative SD of 4.5**, based on MoCA variability in similar populations, where values range from **3.3–3.8** in primary care MCI cohorts. To account for attrition, we include a 20% allowance over the 18-month period, a realistic estimate for this age group. Under these assumptions, our trial will be adequately powered to detect the target treatment effect while maintaining statistical and clinical relevance.

6. Randomisation and Blinding

Randomisation: Eligible participants will be individually randomised 1:1 (TACT+ vs standard care) using a secure, centralised web-based system managed by UCL CCTU. A minimisation algorithm will ensure balance across key prognostic factors: recruitment region, age (<75 vs ≥75), hearing loss severity (PTA 30–40 dB vs 41–60 dB), and living status (alone vs with others).

Allocation concealment and masking: The randomisation sequence will be concealed from nurse assessors. While participants, hearing coaches, and research audiologists cannot be blinded due to the nature of the intervention, all outcome assessments at 12 and 18 months will be conducted by blinded assessors. Participants will be reminded not to disclose their allocation, and assessors will document any suspected unblinding to monitor risk of bias.

7. Internal Pilot and Progression Criteria

The trial will begin with a 6-month internal pilot (Months 25–30) to test recruitment and operations. The goal is to enrol 80 participants across 5–10 early-activated PCN hubs.

Progression will follow pre-specified traffic-light criteria, formally reviewed by the Trial Steering Committee (TSC) and NIHR at Month 30:

Colour	Criteria	Action
Green	80 recruited (100% target)	Proceed without modification
Amber	64–79 recruited (80–99%)	Proceed with modifications and added support, subject to TSC approval
Red	<64 recruited (<80%)	Pause trial; reassess feasibility; TSC/funder decision required

Participants recruited during the internal pilot will be included in the full intention-to-treat analysis. No interim efficacy analysis will be performed.

8. Analysis Plan

8.1 Primary Outcome Analysis (Cognitive Effectiveness)

The primary analysis will follow the intention-to-treat principle, including all randomised participants, regardless of adherence. A **linear mixed-effects model** will estimate the between-group difference in **MoCA score at 18 months**, adjusting for baseline score, site, age, and other confounders. Fixed effects will include treatment, timepoint (12, 18 months), and group×time interaction; a random intercept will model within-subject correlation. This allows efficient use of both follow-up points to estimate treatment effect at 18 months. The primary comparison of interest is the adjusted mean difference in MoCA at 18 months between TACT+ and control, reported with 95% CI and p-value.

8.2 Secondary Outcomes and Subgroup Analyses

Continuous secondary outcomes (e.g. GDS-15, HHIE-S) will be analysed using similar linear or generalised mixed models (with appropriate distributional assumptions or transformations checked). Binary outcomes (e.g. incident dementia at 18 months) will use logistic regression. Exploratory subgroup analyses will examine potential moderators of the effect of TACT+ on outcome, such as baseline hearing loss or cognitive score. No adjustment for multiple comparisons is planned for secondaries given their pragmatic exploratory nature, but the primary outcome analysis will be clearly distinguished. All tests will be two-tailed ($\alpha=0.05$).

8.3 Missing Data Handling

Missing data will be minimised through robust follow-up protocols including multiple contact attempts and use of proxy informants for certain questionnaires if participants permit. If loss exceeds trivial levels, we will apply **multiple imputation** under missing-at-random assumptions, with **sensitivity analyses** (e.g. worst-case, pattern-mixture models) to assess robustness. A detailed Statistical Analysis Plan will be finalised prior to database lock and adhere to CONSORT guidance [53].

9. Health Economic Evaluation

9.1 Within-Trial Cost-Effectiveness Analysis

We will assess cost-effectiveness of TACT+ over 18 months from NHS (direct health and social care costs) and societal (caregiver and out of pocket expenses) perspectives. Resource use will be captured via the CSRI [50] at 6, 12 and 18 months. Intervention delivery costs (staff, devices, app, coach contact) will be logged and costed using NHS reference costs and PSSRU unit costs. Health-related quality of life will be measured using the HUI3 [51,52], a tool that has demonstrated sensitivity to changes in cognitive impairment and hearing loss, at baseline, 6 and 12 and 18 months, enabling **QALY** estimation over the trial duration for each group.

The primary economic analysis will calculate the Incremental Cost-Effectiveness Ratio (ICER)—cost per QALY gained. We will construct cost and outcome totals for each participant and use regression-based methods (to adjust for baseline differences) to estimate mean costs and QALYs by group. Bootstrapping will be used to quantify ICER uncertainty intervals. Results will be presented on a cost-effectiveness plane and summarised with cost-effectiveness acceptability curves at standard willingness-to-pay thresholds (£20k–30k per QALY).

9.2 Long-Term Modelling and Budget Impact

We will build a decision-analytic model to project costs and outcomes beyond 18 months, using trial data and literature on dementia progression. Outputs will include projected dementia incidence, years of independent living, and downstream care costs (like residential care), and the potential long-term benefits of delayed cognitive decline.

A budget impact analysis will estimate national rollout costs, including training, staff and device provision. Sensitivity analyses (deterministic and probabilistic) will explore uncertainty in key drivers of the economic model, such as by varying intervention cost, dementia conversion rates or discount rate. These analyses will identify which parameters most affect cost-effectiveness and test the robustness of our conclusions. Overall, the economic evaluation will inform commissioners about the value for money of implementing TACT+ in routine care.

10. Long-Term Follow-Up via NHS Data Linkage

With consent, trial participants will be linked to NHS records via **NHS Digital Data Access Request Service**. Passive follow-up at 5 and 10 years will capture key long-term outcomes (e.g. dementia diagnosis) without further participant burden. Analyses will follow pre-specified plans, consented by participants, and respecting GDPR and NHS Digital IG requirements.

11. Process Evaluation and Fidelity

Process evaluation methods are detailed in WS4. Key metrics include fidelity, dose, reach, and acceptability. These data will contextualise trial outcomes and support future implementation.

12. Data Management and Quality Assurance

The trial will be sponsored by **North London NHS Foundation Trust (NLFT)** and delivered with the **UCL Comprehensive Clinical Trials Unit (CCTU)**, which will oversee **randomisation, data management, and quality assurance**.

Data will be entered into electronic case report forms (eCRFs) and stored in a secure, GDPR-compliant database with audit trails. Data will be pseudonymised using unique IDs and accessible only to authorised staff. A full Data Management Plan will govern data handling, quality control, monitoring, and archiving.

CCTU will apply risk-based data monitoring, with automated queries to sites to resolve inconsistencies. All adverse events (AEs/SAEs) will be reported per Good Clinical Practice (GCP), with any SAE potentially linked to the intervention escalated to the sponsor. All site staff will be trained in GCP and study-specific procedures. Consent will cover access to medical records and pseudonymised data sharing. REC, HRA, and local R&D approvals will be obtained before launch.

Data will be securely backed up and retained for 10 years. Anonymised datasets will be deposited in an approved repository, following NIHR open data policy. Governance structures are further detailed in the Project Management section.

WS4: Implementation and Process Evaluation
Months 1-63
Milestone 5 & 6: Fidelity report & NHS-ready package - protocols, training and commissioning tools

WS4 embeds implementation science from day 1, mapping barriers to sustained use and NHS scale-up, in line with MRC guidance that implementation should be considered from the outset of intervention development [57].

We will apply the Exploration, Preparation, Implementation, Sustainment (EPIS) framework throughout, attending to both stage and context: from policy (outer context) to organisational readiness (inner context), innovation features, and bridging factors such as training and liaison roles [58]

Process evaluation will run alongside WS1–WS3 and feed directly into WS4. We will track Proctor outcomes—reach (including adoption), acceptability (appropriateness and feasibility), fidelity (including dose delivered), cost (as measured in WP3), penetration, and sustainability—through mixed methods combining quantitative thresholds with qualitative insight (considered from a design perspective) ([59,60] and associated tools.

WS4 will run in five phases, aligned to the structure of the overall programme. An Implementation and Process Working Group will be established from the outset to provide strategic oversight. Members will include people with lived experience of hearing loss or cognitive concerns. PPI contributors will be active members of the group, involved in shaping topic guides, interpreting qualitative findings, and co-developing outputs such as training materials and implementation tools. PPI contributors will play an active role in shaping topic guides, interpreting findings, and co-developing implementation tools. This ensures equity and digital inclusion are embedded from start to finish in a way that is both meaningful and manageable for contributors.

4.1 Phase 1 – Co-production of implementation-informed intervention (aligned WS1)

During WS1 co-design, we will identify early barriers and enablers across stakeholders perspectives (patients, carers, clinicians, commissioners). We will understand what implementation activities are necessary, and their determinants at the individual, organisational and systems levels by using the Theoretical Domains Framework [61] and Normalization Process Theory [62], respectively, to prompt questions and inform key intervention and implementation design features—for example, training nurse assessors in cognitive and hearing assessment, and framing coaches' conversations to address concerns about self-fitting hearing aids.

We will also draw on the Unified Theory of Acceptance and Use of Technology (UTAUT) [63] to explore how performance expectancy, effort expectancy, social influence, and facilitating conditions shape app adoption, particularly for older adults with cognitive concerns.

Finally, we will trial implementation outcome measures—NoMAD [64], Acceptability of Intervention Measure (AIM), and Feasibility Intervention Measure (FIM) [65] —to check they are valid and acceptable for our context and aims.

4.2 Phase 2 – Contextual interviews and intervention data (aligned with WS1/WS2)

Implementation (TDF/NPT/EPIS) We will combine implementation-focused interviews with fidelity monitoring to prepare for WS3 delivery. In ~40 semi-structured interviews with staff, commissioners, carers, and participants, we will use topic guides to examine:

- Acceptability and feasibility of the intervention and delivery model
- "Work needed" and "how this got done" from multiple perspectives
- Setting-level barriers/facilitators, using EPIS to examine outer (e.g. local commissioning) and inner (e.g. staffing, routines) contexts
- Inner and outer contexts, drawing on EPIS to identify setting-level challenges or facilitators, and if there is difference in appropriateness or feasibility of the pathway by this
- Use and acceptability of implementation measures (e.g. NoMAD, AIM, FIM)

Sampling will oversample NIHR's Innovations in Clinical Trial Design and Delivery for the Under-served (INCLUDE) priority groups [66] to ensure that equity barriers and potential contextual differences by resource level are surfaced early.

Fidelity (CFIR-guided) To support fidelity in WS3/WS4, we will analyse:

- Structured coach logs and session templates
- Audio recordings of ~10 % of coaching and fitting sessions (with consent)
- Direct observations of assessment procedures

Findings will be mapped to the Consolidated Framework for Implementation Research (CFIR [67]) and used to co-design a practical 10-item fidelity checklist, alongside an updated implementation logic model.

In parallel, we will identify core versus adaptable elements of the intervention, using fidelity data, qualitative input, and implementation theory to inform later measurement of dose and contextual tailoring in WS3.

4.3 Phase 3 – Hybrid process evaluation in WS3 (Mo 25–54)

A mixed-methods process evaluation will run embedded in the main RCT (WS3) across six purposefully sampled trial sites. Sites will be selected using a maximum-variation matrix (region, deprivation, and early recruitment performance) to ensure contextual diversity. This phase will evaluate how and why the intervention works, for whom, and under what conditions.

Implementation outcomes will be assessed using the RE-AIM framework [68]:

- Reach: We will analyse recruitment and app usage metrics to monitor participant engagement and coach activity, and to assess equity of uptake, considering age, ethnicity, and IMD quintile.
- Dose: We will measure the amount of intervention received (core vs adaptable elements as per phase 2) based on coach records, app usage (via built-in logs), and hearing aid use (as captured in WS3 outcome assessments).
- Fidelity: A bespoke 10-item checklist developed in Phase 2 of WS4 will be applied to ~10% of coaching, hearing assessment, and fitting sessions. Key dimensions will include: (i) coaching support of self-fitting, (ii) support for digital inclusion, and (iii) involvement of communication partners.

Qualitative component:
Up to 50 stakeholders (participants, carers, communication partners, coaches, assessors, supervisors) will contribute via semi-structured interviews or focus groups. Topic guides will draw on implementation theory (NPT) but adapt responsively to emerging themes. Thematic analysis will combine inductive and deductive coding.

Phase 3 will yield:

- A dataset linking variation in implementation (reach, dose, fidelity) to trial outcomes
- A matrix of modifiable barriers and enablers to inform future scale-up
- Fidelity scores and coaching activity metrics to support interpretation of effectiveness variation

4.4 Phase 4 – Implementation toolkit pilot (Mo 50–63)

Following the RCT, we will enter the Implementation and Sustainment phase of the EPIS framework. Despite implementation consideration and planning from the outset, some procedures will require adaptation for routine care. To support this, we will pilot real-world delivery of TACT+ in two diverse PCNs following the end of their trial data selection, and selected for contrasting contexts.

Co-design workshops with local staff, Integrated Care Systems (ICS) leads, and PPIE contributors will translate learning from WS1–3 into NHS-ready resources. This work will map identified barriers into specific behaviour change techniques with practical toolkit resources, again drawing on TDF an NPT and considering inner and outer context.

We anticipate the outputs of the toolkit to include: a CPD-accredited e-learning module, fidelity and equity checklists for ongoing monitoring, an economic impact calculator, and a commissioning brief aligned with ICS priorities and GIRFT[32]. Where feasible, we will assemble a DTAC evidence pack (clinical safety, cybersecurity, interoperability) to support digital adoption.

Toolkit components will be iteratively tested for usability and cost within these pilot PCNs. CPD module acceptability and learning will be assessed with brief structured

quizzes. We will administer the Program Sustainability Assessment Tool (PSAT) to quantify likelihood of sustained delivery.

The toolkit will then be finalised by incorporating feedback from pilot sites and end-of-programme data. Deliverables include the finalised Toolkit v1.0, a sustainability plan, and an implementation report to support NHS commissioning. This phase completes the final step needed to position TACT+ for real-world impact.

WS5: Dissemination and Impact (Months 36-63; Milestone 7: Dissemination outputs and stakeholder engagement report delivered)

Workstream 5 runs from Months 36–63 and focuses on dissemination and stakeholder impact. It is fully described in **Knowledge mobilisation, dissemination and impact** section.

End of Programme Activity
Activity concludes by Month 63. The Chief Investigator (0.2 FTE) will remain in post during Months 64–66 to finalise the NIHR report, and respond to funder queries.

Working in Partnership with Those Affected by Our Research

The TACT+ programme is built on partnership with people living with MCI and hearing loss, and those who support them. From the outset, our approach has gone beyond consultation to integrated co-production, with lived-experience partners embedded in design, delivery, and oversight.

Who has shaped this proposal and why?

The TACT+ programme builds on a robust foundation of public partnership. Our pilot trial was co-designed with three PPIE contributors with lived experience of hearing loss and either personal cognitive concerns or experience caring for someone with dementia, and who were instrumental in designing an acceptable and feasible intervention. Building on this foundation, our current public partners have identified key research questions and defined successful outcomes. Notably, their input established that a 1.5-point difference on the MoCA represents a clinically meaningful slowing of cognitive decline, significantly impacting daily life. Our two PPI co-applicants, Linda Kirby and Terry Emery, have led the Stage 2 application, ensuring our work continues to reflect real-world priorities.

Who will be involved throughout the research lifecycle and how?

We have built in public involvement across all levels:

- **Leadership and oversight**: Our PPI co-applicants are core members of the co-investigator and management team and co-lead work on participant materials and dissemination. An independent public member will sit on the Programme Steering Committee to ensure accountability.

- **Operational co-production**: In WS1, people with lived experience will join mixed co-design workshops to shape the intervention. We anticipate a standing co-production group of 8–10 people, including underserved voices, will meet quarterly to advise on trial materials (WS2/3), interpret findings, co-develop outputs such as WS4 implementation toolkit and help dissemination.
- **Support and inclusion**: All public contributors will receive clear role descriptions, training, and fair reimbursement aligned with NIHR guidance. We will provide flexible and accessible formats, including live captions, Easy Read documents, and hybrid participation options.

Measuring PPIE impact

We will use the Public Involvement Impact Assessment Framework (PiiAF)[69] to develop a logic model of expected PPIE impacts, including improved trial accessibility, relevance of findings, and participant-facing materials. This model will guide reflection and improvement.

Commitment to diversity

We will recruit from high-deprivation areas and ethnic minority communities using NIHR Research Inclusion tools. Flexible modes of involvement—written, spoken, remote, or in-person—will ensure a wide range of voices are heard and included.

Inclusive Research

TACT+ embeds inclusivity across design, delivery, and dissemination, guided by the NIHR INCLUDE framework [66], is embedded throughout the TACT+ programme. Our goal is to reach underserved groups disproportionately affected by hearing loss and at higher risk of dementia, and less likely to access care.

Inclusive design: PPI partners co-develop all participant materials to ensure plain English readability and cultural sensitivity.

Inclusive recruitment: TACT+ targets high-need populations using recruitment in areas of socioeconomic deprivation. Recruitment will target areas of deprivation through GP mail-outs, SMS reminders and opportunistic in-practice invitations to maximise reach into high-deprivation IMD deciles and low-literacy groups.

Removing barriers to participation: To reduce digital exclusion, participants will be offered 4G-enabled tablets (n=278, £98,620) and tech support from hearing coaches as well as flexible onboarding to accommodate fatigue and mobility limitations.

Inclusive methods and analysis: We will monitor recruitment by ethnicity, IMD, and age. Qualitative process evaluation will purposively sample diverse voices, and subgroup analyses (by age, deprivation, ethnicity, and hearing-loss severity) will explore differential impacts and support inclusive implementation.

Accessible dissemination: Findings will be shared via Easy-Read summaries, short videos, and local engagement events. National partners like RNID will help reach underserved communities.

Our Research Team

TACT+ addresses a complex, real-world challenge at the intersection of dementia, hearing care, and digital health. Our team brings together the multidisciplinary expertise needed to tackle it: from clinical trials to audiology, primary care, behavioural science, and co-production.

Prof Sergi Costafreda Gonzalez (CI), Professor of Psychiatry of Ageing at UCL and Consultant at Islington Memory Service, leads the team. He also led the TACT pilot, showing that supported hearing aid use is feasible for people with MCI, and directs the ENHANCE programme on dementia prevention in healthy older adults.

Prof Gill Livingston, co-author of the Lancet Commission on Dementia Prevention, contributes expertise in intervention design and trials, and will act as the NIHR Academy contact. Prof Ross Dunne, consultant psychiatrist and lead author of the Manchester MCI Consensus, strengthens our diagnostic expertise.

We've built strong bridges with audiology and primary care. Dr Devina Maru (RCGP Champion for Hearing Loss) brings frontline primary care insight. Prof Anne Schilder, Director of the NIHR UCLH BRC Hearing Theme, contributes trial and neuroscience expertise. Gareth Smith (Mid and South Essex NHS, former BAA President) and Prof Nishchay Mehta (Tympa rollout lead) ensure our hearing care pathway is practical and safe. Helen Henshaw brings deep expertise in digital auditory interventions. Liz Steed, chartered health psychologist, contributes behaviour change and fidelity design expertise.

The UCL Comprehensive Clinical Trials Unit provides operational and methodological rigour, with Prof Nick Freemantle overseeing design and statistics, and Felicia Ikeji leading delivery. The economic evaluation is led by Monica Panca, expert in ageing and health inequality.

Equity and inclusion are embedded throughout. Fahima Khatun leads our PPIE strategy, with strong links to underserved groups. Dr Sarah Semeraro represents our RNID partnership. Our public co-applicants, Linda Kirby and Terry Emery, guide the programme with lived experience of hearing loss and cognitive concerns.

Our team blends national leadership, practical delivery, and real-world insight—ready to translate innovation into NHS impact.

Research Capacity Building

We will invest 19% of the budget to develop future research leaders in hearing and dementia, targeting key gaps in behavioural science, audiology, and digital health. All posts will be openly recruited in line with NIHR principles. Prof Gill Livingston will serve as the Named Point of Contact for Training and Development, with support

from Prof Costafred, Chief Investigator and who is also the NIHR IAT lead at UCL for psychiatry clinical-academic training.

We will support clinical academic pathways at pre-doctoral and doctoral level, with a focus on under-represented fields. All trainees will be eligible for NIHR Academy membership, with access to mentorship, leadership training, conference support, and networks.

Posts will be advertised in Year 1. The Clinical Audiology RA and PhD 1 will start by Month 6; PhD 2 and the Qualitative RA by Month 12.

Funded capacity posts

PhD 1: Behavioural Science & Digital Health (UCL/QMUL) — WS1 & WS4
Examines how TACT+ supports habit formation via coaching and app use. Supervised by Prof Costafreda, Dr Steed, and Dr Maru, within Behaviour Change, Primary Care and Psychiatry environments.

PhD 2: Audiological Fidelity & Tech (Nottingham/UCL) — WS2 & WS4
Anticipated tp be an NIHR Nottingham BRC audiologist supervised by Dr Henshaw (Nottingham and Prof Mehta (UCL), focusing on fidelity and technical adequacy of supported self-fitting to inform NHSimplementation.

Pre-Doctoral Researcher: Clinical Academic Audiology (UCL) — WS2 & WS3
Serves as trial research audiologist supporting intervention delivery and fidelity. The post offers a pathway to a doctoral fellowship and addresses clinical-academic audiology workforce gaps.

Pre-Doctoral RA: Qualitative & Implementation (UCL) — WS4
Supports process evaluation across workstreams, including stakeholder interviews and toolkit co-development. The RA will be trained in TDF and NPT methods.

All posts align with NIHR Academy standards. PhDs are fully funded by the award and embedded in WS1–4. Pre-doctoral researchers follow formal training plans with protected time (10 days/year), mentoring, and access to methods workshops and NIHR Academy activities. Audiology RAs will also co-author outputs and are encouraged to register for M-level credits or NIHR Associate PI schemes.

****Project management****

TACT+ will be delivered through a three-tier governance structure to ensure scientific rigour, financial accountability, and participant safety across workstreams.

The Programme Management Group (PMG) drives day-to-day delivery, tracking milestones, risk log and finance report and issuing an extraordinary review if any task slips by >30 days. The PMG will double as Trial Management Group (TMG) during WS3 to manage the day-to-day conduct and cross-site delivery of the trial, and will then also include CCTU representatives. A Data Monitoring and Ethics Committee (DMEC) will review unblinded data periodically, focusing on recruitment,

retention, fidelity, and emerging safety issues. Although no interim efficacy analysis is planned (due to low intervention risk), the DMEC will monitor adverse events and recommend action if needed.

An independent Programme Steering Committee (PSC) provides six-monthly scientific and ethical assurance, while a chartered Data Monitoring & Ethics Committee (DMEC) oversees participant safety and data integrity during the trial.

Tier	Composition	Meeting frequency	Core duties
Programme Steering Committee	Independent Chair, statistician, PPI contributor, NIHR observer	Six-monthly	Provide independent scientific and ethical oversight; receive escalations
Data Monitoring & Ethics Committee	Independent statistician, clinician, PPI	Per charter	Monitor trial safety and data integrity
Programme Management Group	CI, Programme Manager, work-stream leads, Trial Manager, PPI co-I, CCTU lead	Monthly to Quarterly (depending on stage of Programme)	Monitor milestones, budget, risk and equity/PPIE metrics
Work-stream teams (WS1–WS5)	Work-stream lead + site representatives	Monthly to Quarterly (while active)	Day-to-day delivery; feed KPIs to PMG

****Knowledge mobilisation, dissemination and impact****

Workstream 5 (WS5) focuses on Dissemination and Impact and is described here to avoid duplication. This workstream ensures that TACT+ findings, tools, and protocols are rapidly translated into real-world impact. Dissemination begins at Month 36 (end of RCT recruitment, when 12-month data are available) and goes to Month 63.

We will target three key audiences: the public, clinicians, and policy/service leads, delivering practical, high-utility resources to support spread and adoption.

Public and Patient Engagement
Working with national charities (e.g. RNID, co-applicant Dr Hannah Semeraro) and our PPIE co-investigators, we will produce accessible outputs including Easy-Read summaries, subtitled “What TACT+ Involves” videos, and quarterly bulletins. Dissemination through the FutureNHS Dementia Hub and partner networks will prioritise the needs of underserved groups identified via the NIHR INCLUDE framework [66].

Clinical Dissemination
Findings will be published in open-access, high-impact journals and presented at key meetings including AAIC, BSA, and RCGP. All reporting will follow CONSORT-EHEALTH to maximise transparency, reproducibility and reusability of digital health

trial data. A CPD-accredited e-learning module (from WS4) will support adoption in memory, audiology, and primary care services. We will also run clinical workshops to engage front-line professionals.

Policy, Service & Commissioning Leadership
We will engage leadership stakeholders with an NHS-ready TACT+ Implementation Toolkit, grounded in implementation science and co-design, to include:

- A commissioning brief aligned with ICS priorities and GIRFT[32]
- A CPD-accredited training module
- Fidelity and equity monitoring checklists
- A costing template to support local decisions
- (Resources permitting) a DTAC evidence bundle

Resources will be hosted on a platform such as the NIHR ARC Implementation Exchange. We will collect feedback from early adopters and track toolkit use to inform adaptation and spread.

****Intellectual property and commercialisation****

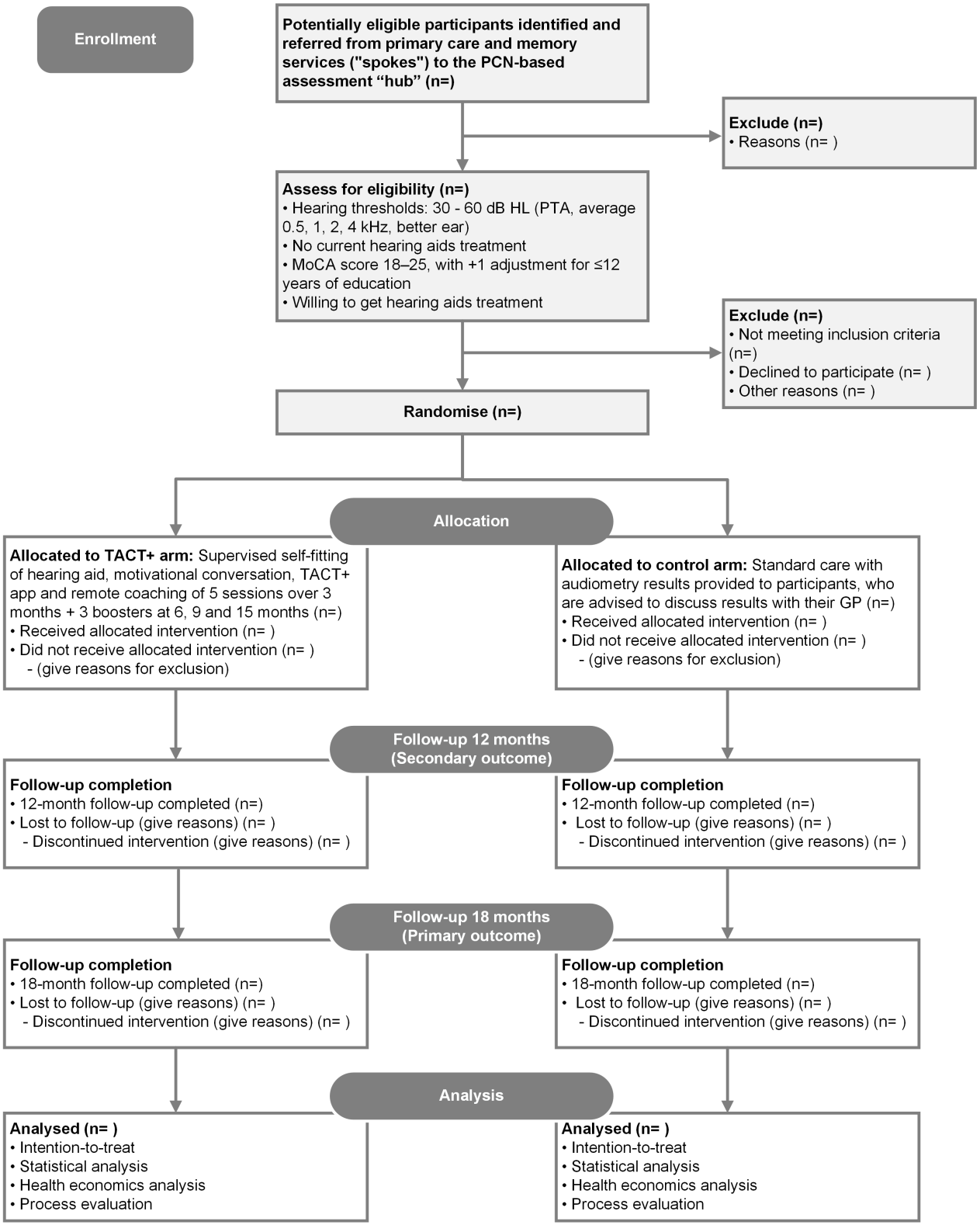
Background IP – NLFT is the sponsor of both ENHANCE and TACT+, owns the ENHANCE code-base on which the TACT+ App will be base, and has granted the project an unrestricted, royalty-free licence; no third-party restrictions apply (Schedule B). We will seek relevant permissions for trial instruments.

Foreground IP – All new source code, algorithms, content and training materials created during TACT+ will be automatically assigned to NLFT.

Collaboration agreements – TACT+ Contractors are expected to be Glitchers and TympaHealth (subject to UCL contracting processes) and will operate under work-for-hire sub-contracts that (i) assign all IP to NLFT, (ii) waive moral rights, and (iii) prohibit independent commercial use without NLFT/NIHR approval. Draft agreements will be supplied to NIHR before signature.

Protection & management – NLFT will maintain an IP register, review quarterly w ith UCL Business (UCLB), and decide whether to file design-right or patent applications for novel components.

Exploitation pathway – The primary route is a non-exclusive, royalty-free licence for NHS deployment via an open-source compliant model. Towards the end of the project, NLFT/UCLB will explore a social-enterprise spin-out or SaaS licensing to international providers. Any commercial licence will preserve research & NHS rights, favour NHS uptake at preferential or no cost, and, if commercial exploitation is envisaged, a revenue sharing agreement will be decided at that point, with advice from UCLB who has expertise and has a longstanding collaboration advising NLFT-sponsored projects in these matters.


Enrollment
Potentially eligible participants identified and referred from primary care and memory services ("spokes") to the PCN-based assessment "hub" (n=)
Exclude (n=)
• Reasons (n=)
Assess for eligibility (n=)
• Hearing thresholds: 30 - 60 dB HL (PTA, average 0.5, 1, 2, 4 kHz, better ear)
• No current hearing aids treatment
• MoCA score 18–25, with +1 adjustment for ≤12 years of education
• Willing to get hearing aids treatment
Exclude (n=)
• Not meeting inclusion criteria (n=)
• Declined to participate (n=)
• Other reasons (n=)
Randomise (n=)
Allocation
Allocated to TACT+ arm: Supervised self-fitting of hearing aid, motivational conversation, TACT+ app and remote coaching of 5 sessions over 3 months + 3 boosters at 6, 9 and 15 months (n=)
• Received allocated intervention (n=)
• Did not receive allocated intervention (n=)
- (give reasons for exclusion)
Allocated to control arm: Standard care with audiometry results provided to participants, who are advised to discuss results with their GP (n=)
• Received allocated intervention (n=)
• Did not receive allocated intervention (n=)
- (give reasons for exclusion)
Follow-up 12 months (Secondary outcome)
Follow-up completion
• 12-month follow-up completed (n=)
• Lost to follow-up (give reasons) (n=)
- Discontinued intervention (give reasons) (n=)
Follow-up completion
• 12-month follow-up completed (n=)
• Lost to follow-up (give reasons) (n=)
- Discontinued intervention (give reasons) (n=)
Follow-up 18 months (Primary outcome)
Follow-up completion
• 18-month follow-up completed (n=)
• Lost to follow-up (give reasons) (n=)
- Discontinued intervention (give reasons) (n=)
Follow-up completion
• 18-month follow-up completed (n=)
• Lost to follow-up (give reasons) (n=)
- Discontinued intervention (give reasons) (n=)
Analysis
Analysed (n=)
• Intention-to-treat
• Statistical analysis
• Health economics analysis
• Process evaluation
Analysed (n=)
• Intention-to-treat
• Statistical analysis
• Health economics analysis
• Process evaluation

Logic Model: Treating auditory impairment to preserve cognition in people with mild cognitive impairment (TACT+)

The problem

People with MCI have increased risk of dementia.

Many also have untreated hearing loss, which exacerbates isolation, depression, reduced quality of life, and may accelerate cognitive decline.

MCI patients face barriers accessing hearing care, leading to low hearing aids uptake.

Behavioural and psychosocial barriers reduce uptake and use of hearing aids

Identification of MCI & hearing loss and subsequent care pathway for missing from primary care.

Primary care staff untrained in hearing care provision

Intervention Targets

Establish an integrated hearing care pathway within PCN hubs for people with MCI.

Increase identification of MCI and hearing loss

Improved access to hearing aids.

Capability and motivation to use hearing aid addressed

Continuity and adherence to hearing care supported

Trained primary care workforce.

Intervention Components

Nurse assessment and explanation of hearing and cognitive findings

Explanation of link between cognition & hearing loss

Supported self-fitting hearing aids with guidance (hearing coaches)
Intro TACT + app-based games: Reminders, instructional videos, auditory rehabilitation games.

Digital inclusion support: including provision of tablets and connectivity

Scheduled remote check-ins with coach: Coach explores barriers, Troubleshooting, prompting, reinforcement.

Coach-led reactive support: Messaging, troubleshooting via **TACT +** app

Coach supervision by Audiologist: Remote and answers queries

Mechanisms of Action

TACT+ app's instructional videos and guides → ↑ awareness on the hearing – cognition link → ↑ belief in need hearing care.

Capability and skills (acclimatisation)

Motivation to use hearing aid

Repeated prompts, remote coaching, and troubleshooting support → ↑ recall to use and habit formation.

Engaging auditory rehabilitation games → ↑ acclimatisation

Better adherence to hearing aid use → ↓ cognitive load for hearing → ↑ available cognitive resources for other tasks.
Easier social engagement → ↓ isolation, ↑ mood and support.

Timely and continuous care → ↓ fatigue, ↑ trust in services.

Outcomes

SHORT TERM
↑ Hearing aid uptake and consistent use
↑ Self-reported hearing ability
↓ Perceived effort/fatigue during communication

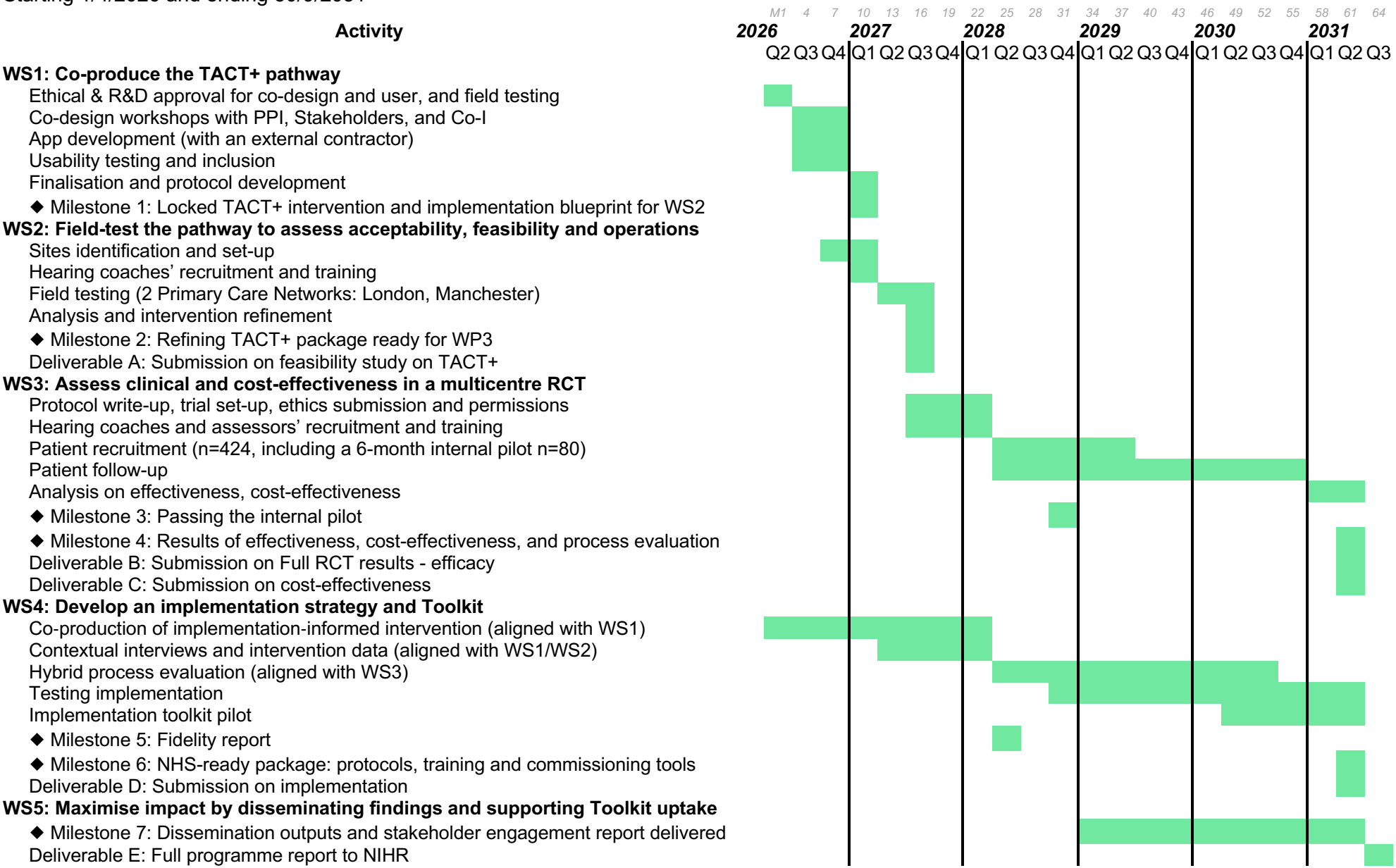
LONG-TERM
↑ Hearing (IOI-HA, HHIE/HHIA)
↑ Quality of life (HUI3)
↑ Social functioning (LSNS)
↑ Mood (GDS-15)
↓ Cognitive decline (MoCA)
↓ Dementia conversion rate

Capacity Building in Audiology and Mental Health Services

GDS-15=Geriatric Depression Scale – short form; HHIA/HHIE = Hearing Handicap Inventory for the Adult/Elderly; HUI3 = Health Utilities Index Mark 3; IOI-HA = Outcome Inventory for Hearing Aids; LSNS = Lubben Social Network Scale; MCI=Mild Cognitive Impairment. MoCA, Montreal Cognitive Assessment; PCN, Primary Care Network.

GANTT chart: Treating auditory impairment to preserve cognition in people with mild cognitive impairment (TACT+)

Starting 1/4/2026 and ending 30/9/2031



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